VR-Forms

An Operational Two-Register Apparatus over VR-Sets

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Abstract

VR-Forms is the fourth work in the cycle developing the VR programme (Reznik 2026, preprints VR. A Formal System, VR-Numbers, VR-Sets). It introduces a two-register apparatus on top of VR-Sets: an operational register (operational sets of VR-Sets, with closure principle and reference-based $\in$) and a formal register (syntactic descriptions without operational commitment). The formal register provides a universal language for everything that is not operationally realised — classical uncountable objects, paradoxical classes, mythological and theological terms, philosophical categories. The two registers are connected by a conservativity theorem (Theorem III.1): the formal register is a conservative extension of the operational, generating no new operational commitments and proving nothing in the operational register that VR-Sets could not already prove. The transit rule allows formal reasoning over uncountable intermediaries, with results automatically translating into the operational register when the conclusion is itself operational. The apparatus closes the operational contour of the VR cycle: VR can now speak about everything, while ascribing no ontology — only operations, $\varnothing$ itself the base. Classical paradoxes (Russell, Vitali, Skolem) and classical uncountable objects ($\mathbb{R}$, $\wp(\mathbb{N})$, proper classes) receive a clear status — formal terms without operational correlate. The same status, by uniform application of the principle of forms, extends to mythological, theological, and literary descriptions, demonstrating the universality of the formal register beyond mathematics.

Changes in version 1.0.1. A new Part IX has been added: "Lean 4 Formalisation of VR-Forms: Methodological Observations." This part documents the Lean 4 formalisation of the formalisable core of the apparatus — Parts II–IV (formal language, realisability, transit) plus mixed formulas of §VII.2 (Reznik 2026, Lean VR-Forms, Software DOI 10.5281/zenodo.20355757; Git tag v1.3-vr-forms). The formalisation is a shallow embedding over mathlib with one explicit structural boundary at conservativity (Theorem III.1), which would require deep-embedded Formula and Derivation types beyond the scope of this cycle. Ten methodological observations are reported in four thematic clusters: foundation-level properties (§IX.1, two observations), the central boundary at conservativity and its four substructural manifestations (§IX.2, five observations), structural patterns in the apparatus (§IX.3, two observations), and cross-cycle integration (§IX.4, two observations including the finale on zero Classical.choice usage). The Lean formalisation contains 18 public objects across 5 files; three are axiom-free (Stage 1), fifteen sit at [propext, Quot.sound], and **none requires Classical.choice** — making it the most axiom-minimal of the four VR Lean cycles, a stricter result than the original cycle plan predicted. §I.8 (Structure of the work) is updated to nine parts; the references and acknowledgement are extended to reflect the parent-child review architecture used in the Lean formalisation, identical to that used in VR-Sets v1.0.1. No mathematical content of Parts I–VIII has been revised; the changes are additive (Part IX is new) and editorial.

Companion publications. Preprint VR-Sets v1.0.1 (DOI 10.5281/zenodo.20354628) similarly documents the Lean 4 formalisation of its mathematical core in Part X. The full VR cycle in its v1.0.1 form consists of four preprints (VR, VR-Numbers, VR-Sets, VR-Forms) and four Lean formalisations, all available on Zenodo with cross-references.

Keywords: operational foundation, two-register logic, formal terms, conservative extension, transit pattern, Russell paradox, Skolem paradox, Vitali set, AFA, non-well-founded sets, nominalism, universal formal language, Lean 4, formal verification, mathlib, axiom audit, shallow embedding.

Part I. Foundations

I.1. Place in the cycle

VR-Forms is the fourth work in the cycle developing the **VR** programme:

VR. A Formal System (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20212092) — a minimal axiomatisation of arithmetic on three primitives {∅, →, t} and four axioms; equivalent to Peano arithmetic.

VR-Numbers (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20272743) — operational extensions over the natural numbers of VR: ℤ, ℚ, ℝ, ℂ as operational procedures, without introducing pairs as objects.

VR-Sets (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20303536) — an operational theory of sets: a set is a describable functionality, ∈ is reference rather than containment. Russell's paradox dissolves operationally. Countability of the universe as a consequence of operationality.

VR-Forms is the present work. It introduces a second register — **formal terms**, descriptions without operational force. These terms are formulas, and nothing more. They serve as a language in which VR can speak about everything that it does not operationally realise: about classical uncountable objects, paradoxical classes, figures of myth and literature — about all descriptions that cannot be reduced to operational action.

Substantive connection with the preceding works: VR-Sets opened the boundary of the operational. VR-Forms describes how to speak about what lies beyond this boundary, without crossing it and without blurring it.

I.2. The initial position

The slogan of the whole cycle: *«nothing is — all is doing»*. Applied to arithmetic, it gave VR; to numbers, VR-Numbers; to sets, VR-Sets. In all three works the slogan operates at the foundational level: nothing simply is — ∅ itself is a nullary operation; everything is operational action upon ∅ and upon the results of prior actions.

VR-Forms adds a second mode. Beside what acts (operational functionalities, with ∅ the base operation), there is introduced what is said. Not as a new entity of any kind — that would violate minimalism. Rather as a separate syntactic register, carrying no operational commitments.

The full formulation of the position:

«Nothing simply is. Everything either acts or is said. What acts is operational, with ∅ the base operation; what is merely said is formal and is backed by no operation.»

That is VR-Forms in a single sentence.

I.3. Two registers

We introduce two registers of reasoning. They differ not in subject matter and not in language, but in whether a term is backed by an operation — whether something can actually be carried out behind it. The label "operational register" below is technical: it names the register of terms that VR operationally realises, and implies no metaphysical claim — the cycle ascribes no ontology, $\varnothing$ itself being a nullary operation.

The operational register

This is the register of VR-Sets. The following principles hold:

— ontologically, nothing simply is: $\varnothing$ is the nullary base operation;

— all other terms are operational functionalities, described by a finite procedure;

— the closure principle: every describable operational functionality is a set;

— contradictory descriptions do not specify functionalities and therefore do not specify sets.

In this register terms have reference: they refer to operational entities — operations upon $\varnothing$ and upon the results of prior operations ($\varnothing$ itself being the nullary operation, not a being).

The formal register

This is the new register introduced by the present work. The following principles hold:

— terms are **formulas**;

— a formula specifies a **formal term**, and that is all it specifies;

— no formula is required to have reference;

— a formal term is a formal term and does not presuppose any other entity behind it.

In this register terms are speech. They refer to nothing by necessity. They are form — and only form.

Passage between registers

The same term may be considered in both registers. The description «the set of all functions $\mathbb{N} \to \mathbb{N}$» in the operational register requires verification: is it operational? Answer: no, the general set of all functions is not computable; hence in the operational register it does not specify a set. The same description in the formal register is a formal term; no verification is required.

Passage between the registers is not a metaphysical act. It is a passage between two ways of considering one and the same syntax. In one case we ask «is there a corresponding entity?»; in the other, we do not ask.

I.4. What a formal term is

A formal term is the syntactic record of a description. No more.

A formal term should not be thought of as a «potential set», or a «classical set to which VR has no access», or a «shadow object». All such readings are reifications of a formula, attempts to ascribe reference to it. In VR-Forms this attempt is rejected as a matter of principle.

A formal term is form without operation. A record that describes — but no operation carries the description out; in the formal register the question of being is not posed at all.

This is not agnosticism. We do not say «perhaps it is, perhaps it is not». The question of being does not arise at all — nothing simply is. The registers differ only in whether an operation stands behind the term; to ask after the being of a formula is a category mistake. The proper question — is there an operation that carries the term out? — is asked in the operational register. There we obtain an answer: either «a corresponding operational functionality can be carried out» (and then the formal term corresponds to a VR-Sets set), or «not» (and then the formal term remains purely formal and has no operational correlate).

I.5. Examples of formal terms

Examples are divided into three groups, demonstrating the universality of the formal register.

Group 1. Mathematical formal terms

Classical ℘(ℕ) — the description «the set of all subsets of ℕ». In the operational register of VR-Sets this is not a set in the full classical sense: VR contains a countable number of describable subsets of ℕ. In the formal register «℘(ℕ)» is a formal term. The statement «the classical cardinality of ℘(ℕ) is $2^{\aleph_0}$» belongs to the formal register and has no operational force.

Classical ℝ — the description «the set of all real numbers». In the operational register ℝ_VR is the countable set of computable reals. In the formal register «classical ℝ» is a formal term distinct from ℝ_VR.

Russell's class — the description {x : x ∉ x}. In the operational register it is contradictory and does not specify a set (VR-Sets §II.3). In the formal register R = {x : x ∉ x} is a formal term. Russell's theorem now takes the form: R is a formal term having no operational correlate in VR-Sets.

The Vitali set — the description «a set of representatives of equivalence classes on ℝ under the relation $x - y \in \mathbb{Q}$». Requires choice over an uncountable family. Not operational. In the formal register a formal term.

Group 2. Paradoxical formal terms

The liar — the description «a statement equivalent to its own negation». In VR-Sets §VII.2 it was shown that in the operational register it has no truth value. In the formal register the liar is a formal term; paradoxicality is a property of form, not an operational failure.

The set of all sets — the description «an object whose elements are all sets». In the operational register contradictory. In the formal register a formal term.

Group 3. Non-mathematical formal terms

The apparatus of the formal register is not confined to mathematics.

A dragon — the description «a fire-breathing lizard-like creature of large size with wings». This is a formula, a description. In the operational register of VR-Sets — no operational functionality over $\varnothing$. In the formal register — a formal term.

God in monotheistic understanding — the description «the unique omnipotent eternal creator of the world». A formula. A formal term.

Unicorn. Phoenix. Centaur. Minotaur. Leviathan. All mythological beings are formal terms.

Soul. Idea in the Platonic sense. Universal. Essence. All classical philosophical terms are formal terms.

This last point has a fundamental significance. **The formal register of VR-Forms covers speech without exception.** Everything that is spoken of but is not operational is a formal term.

This is not reductionism. We do not say that dragons do not exist, or that God does not exist, or that universals do not exist. We say only: in the operational universe of VR they are not; in the formal register one can speak of them as formal terms. The question of their existence outside the operational universe of VR is not a mathematical question and not a question of the present work.

I.6. What VR-Forms does with speech

The introduction of the formal register changes the status of classical contested questions.

The question of the reality of the uncountable. In the operational universe of VR the uncountable is not — the operational universe is countable (VR-Sets §VI.1). Classical mathematics freely uses the uncountable. VR-Forms gives a precise formulation: classical $\mathbb{R}$ is a formal term. Statements of classical mathematics about $\mathbb{R}$ are statements of the formal register. The dispute about the «reality» of the uncountable is a dispute about whether the corresponding formal term has an operational correlate beyond VR. Mathematics does not answer this question; it shows only that in the operational universe of VR there is none.

The question of paradoxes. Russell, the liar, Berry — all of them show: the corresponding formal terms have no operational correlate in VR. This is the content of paradoxicality: the paradoxical description remains in the formal register. No damage to the operational universe of VR is done.

The question of the reality of non-mathematical entities. Dragons, gods, unicorns. In the operational universe of VR they are not. In the formal register one can speak of them. Whether they exist in some other ontology is not a question of VR-Forms. The work only provides a formal language permitting talk of them without operational commitment.

VR-Forms does not deny the existence of anything. It distinguishes what is operational in its own universe from what is merely said and may (in another ontology) exist otherwise. About the first — operational speech. About the second — formal speech.

I.7. Connection with the philosophical tradition

The position of VR-Forms has precedents in the history of philosophy.

Medieval nominalism (Ockham, Buridan). Universals are names, not entities. Only concrete individuals exist. VR-Forms inherits this position: formal terms are names; only the operational is real. The difference: for Ockham reality is ascribed to material individuals; in VR — only to $\varnothing$ and actions upon it. A more strict nominalism.

Mathematical constructivism (Brouwer, Bishop). Only that which is constructed exists. VR-Forms agrees with this in the operational register. The difference: VR-Forms adds the formal register, in which one may speak of the non-constructive without claiming its existence. Classical constructivism refuses talk of the non-constructive; VR-Forms permits the talk but deprives it of operational force.

Formalism (Hilbert). Mathematics is manipulation of formulas without reference. VR-Forms inherits formalism in the formal register. The difference: VR-Forms does not reduce all mathematics to the formal register. Part of mathematics is operational — this is VR-Sets. Formalism deprives all mathematics of reference; VR-Forms — only its uncountable part.

Fictionalism (Field, Balaguer). Mathematical objects are useful fictions, not existing in the proper sense. VR-Forms agrees with this regarding uncountable and paradoxical objects. The difference: VR-Forms provides a formal apparatus for working with fictions, not only a philosophical position.

Late Wittgenstein. «Whereof one cannot speak, thereof one must be silent» — but be silent in a metaphysical sense, not a linguistic one. One may speak, while being aware that the speech is not a description of reality. VR-Forms continues this line: the formal register is speech aware of its referentlessness.

VR-Forms does not introduce anything unprecedented. It conducts a known philosophical position through the formal apparatus of VR. What is new in it is the combination of no ontology ($\varnothing$ as a nullary operation, not a primitive) with formal openness (any description is a formal term).

I.8. Structure of the work

The work consists of nine parts.

Part I. Foundations. Place in the cycle, initial position, two registers, formal term, examples (including non-mathematical), connection with the philosophical tradition.

Part II. The formal language. Precise definition of a formal term. Syntax of the formal register. Rules of formation of formal terms. Difference between a formal term and an operational set.

Part III. The principle of forms and consistency. The principle parallel to the closure principle of VR-Sets: every description specifies a formal term. Consistency of the extension.

Part IV. Transits. When work in the formal register yields results of the operational register. The transit rule. The notion of witness. Safe and problematic transits.

Part V. Forms of mathematics. Application to mathematical formal terms: $\mathbb{R}$, $\wp(\mathbb{N})$, Vitali, paradoxical classes. What is translated into the operational universe of VR, what remains in the formal register.

Part VI. Forms of speech. Application to non-mathematical formal terms: dragons, gods, philosophical categories. Universality of the formal register.

Part VII. Two-register logic. Logical properties of the extended system. Preservation of classical logic in both registers. Difference in rules of inference.

Part VIII. Open questions. Connection with proof theory. Preparation for VR-Audit. Possible applications of the formal register in other areas.

Part IX. Lean 4 formalisation of VR-Forms: methodological observations. Added in version 1.0.1. Documents the Lean 4 formalisation of the two-register apparatus (Reznik 2026, Lean VR-Forms, Software DOI 10.5281/zenodo.20355757; Git tag v1.3-vr-forms), reporting ten methodological observations from the formalisation grouped into four thematic clusters: foundation-level properties (Part IX §IX.1, two observations on import-freedom and inheritance from VR-Sets Classical-free closure layer); the central structural boundary at conservativity and four substructural observations (Part IX §IX.2, five observations); structural patterns in the apparatus (Part IX §IX.3, two observations); cross-cycle integration (Part IX §IX.4, two observations including the finale on zero Classical.choice usage). The formalisation is the most axiom-minimal of the four VR Lean cycles.

I.9. A note on tone

The work touches both the foundations of mathematics and speech in general. It is mathematical in apparatus and philosophical in scope. The tone is sustained in the manner of the previous works of the cycle: precision of definitions, sparsity of rhetoric, transparency of motivation.

The appearance of «dragons» and «gods» as examples of formal terms is not an artistic device or provocation. It is a direct consequence of the position: the formal register is universal. If it really covers speech without exception, it must be demonstrated on speech without exception.

Part II. The Formal Language

II.0. Introduction

Part I set out the general position: the formal register as speech without operational commitments. The present part translates this position into formal apparatus.

Contents of Part II: precise definition of a formal term; syntax of the formal register; rules of formation of formal terms; the relation of a formal term to an operational set of VR-Sets.

Part II is technically minimal. Its task is to fix the apparatus on which Parts III–VII will build substantive results. No new operational moves are made here; everything rests on Part I.

II.1. Alphabet and terms

The formal language of VR-Forms is built over the same alphabet as VR-Sets, extended by a register-separator.

Alphabet

The alphabet of the formal language contains:

— the symbol $\varnothing$;

— the functional symbol t (successor);

— logical connectives $\to$, $\neg$, $\wedge$, $\vee$, $\leftrightarrow$;

— quantifiers $\forall$, $\exists$;

— the membership relation $\in$;

— the operational identity relation $\equiv$ (from VR-Sets);

— variables x, y, z, …;

— brackets {, }, (,);

— the register-separator symbol $\ulcorner\cdot\urcorner$ (see below).

The alphabet remains finite, so that countability of the syntax is preserved (VR-Sets §VI.1).

Terms and formulas

Terms and formulas are defined inductively in the standard way:

— $\varnothing$ is a term;

— if τ is a term, then $t(\tau)$ is a term;

— variables are terms;

— a description of the form $\{x : \varphi(x)\}$, where φ is a formula with one free variable x, is a term;

— other formulas are built from atomic $x \in y$, $x \equiv y$ by connectives and quantifiers.

So far this is exactly the language of VR-Sets. The difference with VR-Forms arises in the rules of **use** of terms, not in their syntax.

II.2. Register-separator

The same term may figure in the operational or in the formal register. To make the distinction explicit in writing, a syntactic separator is introduced.

Definition II.1 (register-separator)

For any term τ, the notation $\ulcorner\tau\urcorner$ means: the term τ is considered in the formal register. The notation τ without brackets means: the term is considered in the operational register.

$\ulcorner\cdot\urcorner$ is not a function or operation. It is a sign of the **mode of consideration**. It indicates in which of the two registers a given term is read.

Examples

— $\varnothing$ is the empty set in the operational register. By Lemma 2 of VR-Sets it is the unique operational object with empty functionality.

— $\ulcorner\varnothing\urcorner$ is a formal term, the description «of the empty». In the formal register this is simply a formula; no operational commitments. (In practice for $\varnothing$ the operational and formal registers coincide, since the description directly corresponds to an operational object; the separator is here redundant but admissible.)

— $\ulcorner\{x : x \notin x\}\urcorner$ is the Russell formal term. Admissible in the formal register. The corresponding operational term $\{x : x \notin x\}$ does not specify a set (VR-Sets §II.3).

— $\ulcorner\mathbb{R}\urcorner$ is the formal term «classical set of reals». Admissible in the formal register.

— $\mathbb{R}$_VR is the notation for the operational set of computable reals of VR-Sets. In the operational register. Not the same as $\ulcorner\mathbb{R}\urcorner$.

A note on practice of notation

In the work, the separator $\ulcorner\cdot\urcorner$ is used explicitly where the register is otherwise ambiguous. In contexts where the register is clear from the text (for instance, «classical $\mathbb{R}$» — formal register; «$\mathbb{R}$_VR» — operational), the separator is omitted.

II.3. Definition of a formal term

We now give the central definition of the present work.

Definition II.2 (formal term)

A **formal term** is any correctly built syntactic record of a description, considered in the formal register.

Formally: a formal term is a pair (τ, F), where τ is a syntactically correct term of the language of VR-Forms, and F is the register-marker indicating «formal». The notation $\ulcorner\tau\urcorner$ denotes this pair.

Commentary

(1) A formal term is a **pair**: syntax plus indication of mode. The same τ specifies different entities depending on the register: an operational set (in the operational) or a formal term (in the formal).

(2) «Correctly built syntactic record» means: the term is built according to the rules of §II.1. No other conditions — operationality, consistency, constructivity — are imposed. A formal term may be contradictory, non-operational, refer to uncountable collections, describe the impossible. All this is permitted.

(3) In the formal register there is no distinction between «existing» and «non-existing» formal terms. All formal terms are formal terms. The differences in their content (countable, paradoxical, non-mathematical) are properties of **the description**, not properties of being.

II.4. The principle of forms

Corresponding to the closure principle of VR-Sets (every describable operational functionality is a set), there is in VR-Forms a parallel principle pertaining to the formal register.

The principle of forms

Every syntactically correct record of a description specifies a formal term.

No restrictions: operationality is not required, consistency is not required, describability in any special sense is not required. It suffices that the record is syntactically correct by §II.1.

Comparison with the closure principle

The closure principle of VR-Sets applies in the operational register and requires operational describability. Contradictory descriptions (Russell) do not specify sets. This is a **conditional** principle: given operationality — there is a set.

The principle of forms applies in the formal register and requires nothing other than syntactic correctness. This is an **unconditional** principle: every description is a formal term.

Parallel and difference: both principles generate objects of their registers; but in the operational register the objects are operations (actions upon ∅), and in the formal — they are formulas. The difference is not in the statement of the principles, but in the operational standing of the objects they generate.

II.5. Relations and operations on formal terms

Formal terms are formulas. Yet one may work with formulas: formulate statements, derive consequences, compare. This work is work in the formal register.

Membership

The notation $\ulcorner x \in \tau \urcorner$ means: the formal statement «x belongs to the formal term τ». The truth value of this statement is determined by the inference rules of the formal register, not by the operational register of VR-Sets.

Example: let $\tau = \ulcorner\{x : \varphi(x)\}\urcorner$. Then $\ulcorner y \in \tau \urcorner$ is true in the formal register if and only if $\varphi(y)$ is true (in the formal register). This is the standard scheme of comprehension for formal terms.

Comprehension in the formal register is **not restricted** by an axiom of separation. Unlike ZF, where comprehension applies only to elements of already existing sets, in the formal register any description $\{x : \varphi(x)\}$ directly specifies a formal term. This is what makes the Russell formal term a legitimate object of the formal register.

Equality and identity

In the formal register the equality of two formal terms $\ulcorner \tau_1 \urcorner = \ulcorner \tau_2 \urcorner$ is a statement about their **descriptions**. Two formal terms are equal if they are given by one and the same formula (up to renaming of bound variables) or if the equivalence is provable in the formal register.

This differs from ≡ of VR-Sets — the operational identity requiring coincidence of functionalities. In the formal register there are no functionalities; there are formulas.

Logical operations

Connectives and quantifiers apply to formulas of the formal register in the usual way. Classical logic is preserved. No special inference rules for formal terms are introduced.

This is an important technical feature: the formal register is **not a new logic**; it is the same logical apparatus applied to a new type of objects (formal terms instead of operational sets).

II.6. Difference between a formal term and an operational set

Let us gather in one place a systematic comparison.

What is shared

— The same syntax: both formal terms and operational sets are written by formulas of one language.

— The same logic: classical, with the usual rules.

— The same starting primitive: $\varnothing$ is the base object of both registers.

What differs

Register. An operational set is in the operational. A formal term is in the formal.

Operational standing. An operational set is an operation upon $\varnothing$. A formal term is a formula.

Condition of existence. An operational set requires operational describability (closure principle). A formal term requires only syntactic correctness (principle of forms).

Paradoxes. Contradictory descriptions do not specify operational sets. Contradictory descriptions specify formal terms without restriction.

Universe. All operational sets form the countable universe of VR-Sets. All formal terms form a countable syntactic universe (since formulas are finite sequences over a finite alphabet).

Identity. Operational identity $\equiv$ is the coinductive coincidence of functionalities. Formal identity $=$ is syntactic or formal-register-provable coincidence of descriptions.

Connection

To every operational set A there corresponds exactly one formal term $\ulcorner A \urcorner$ — the description of A itself, considered in the formal register. Conversely: not every formal term has an operational correlate. Those that do form a distinguished subfamily of formal terms — let us call it the **operationally realisable** formal terms.

This subfamily is countable (by countability of operational sets). The set of all formal terms is likewise countable (by finiteness of the alphabet). Both registers thus remain within the countable operational universe of VR.

II.7. Operationally realisable formal terms

We introduce this notion precisely — it will be needed in Part IV for the definition of transit.

Definition II.3

A formal term ⌜τ⌝ is called **operationally realisable** if in the operational register there exists an operational set A such that the description τ corresponds to the functionality A.

Equivalently: ⌜τ⌝ is operationally realisable if τ (without the formal-register marker) is an admissible description in VR-Sets satisfying the closure principle.

Examples of realisable formal terms

⌜∅⌝, ⌜{∅}⌝, ⌜ω⌝, ⌜℘_VR(ω)⌝, ⌜ℝ_VR⌝ — all of these are realisable. Each corresponds to an operational set of VR-Sets.

Any description of a computable functionality is an operationally realisable formal term.

Examples of non-realisable formal terms

⌜{x : x ∉ x}⌝ — the Russell term. Not realisable (the description is contradictory).

⌜ℝ⌝ (classical) — not realisable (uncountable, not operationally describable).

⌜the Vitali set⌝ — not realisable (requires choice over an uncountable family).

⌜dragon⌝ — not realisable (no operational action corresponds to the description).

Remark

The distinction «realisable / non-realisable» is a property of the formal term as a pair (description, register). It only tells whether the formal term has an operational correlate. The formal term itself does not become «better» or «worse» from the presence or absence of realisation; in the formal register all terms are equal.

The distinction matters only when we wish to **translate** results of the formal register into the operational — this is the subject of Part IV.

II.8. Connection with proof theory

The two-register system of VR-Forms structurally resembles several constructions known in logic.

Two-sorted logic. A standard extension of first-order logic in which variables are divided into sorts. VR-Forms is formally a two-sorted system, with sorts «operational set» and «formal term». However, the sorts in VR-Forms are not symmetric: the formal sort **includes** the operational (every operational set has a formal correlate).

Conservative extensions. An extension of a theory is called conservative if no new statements about old objects are provable in it. Part III will show: VR-Forms is conservative over VR-Sets in the operational register. The formal register does not allow new statements about operational sets to be proved; it only provides a language for talking about the non-operational.

Realisability (Kleene, Kreisel). A notion linking constructive mathematics to classical: a formula is considered «realisable» if it has a constructive witness. The notion of operational realisability in §II.7 is an analogue of realisability specialised to the operational universe of VR.

Internal set theory (Nelson). An extension of ZFC separating standard and nonstandard objects. Structurally similar, but the motivation is opposite: IST introduces the nonstandard as a tool for proofs about the standard. VR-Forms introduces the formal as a way to speak about the non-existent without claiming proofs.

A detailed comparison with these systems — Part VII.

II.9. Summary of Part II

Established:

(1) Alphabet and syntax of VR-Forms coincide with VR-Sets, with the addition of the separator $\ulcorner\cdot\urcorner$ for indicating the register (§II.1, §II.2).

(2) A formal term is a pair (syntactic record, formal register); notation $\ulcorner\tau\urcorner$ (Definition II.2, §II.3).

(3) The principle of forms: every syntactically correct record is a formal term. Unconditional, in contrast with the conditional closure principle (§II.4).

(4) Logic and operations are classical, applied to formal terms in the usual way (§II.5).

(5) Systematic comparison of formal term and operational set (§II.6).

(6) The notion of operationally realisable formal term (Definition II.3, §II.7).

(7) Connections with two-sorted logic, conservative extensions, realisability, IST (§II.8).

What follows. Part III proves consistency of the extension: VR-Forms is conservative over VR-Sets in the operational register. This theorem secures the safety of work in the formal register — no new operational commitments are generated.

Part III. The Principle of Forms and Consistency

III.0. Introduction

Part II provided the apparatus: the formal language, register-separator, definition of a formal term, principle of forms. The present part proves a key property of this apparatus — its **safety**.

Safety means: adding the formal register to VR-Sets does not generate new contradictions and does not allow anything new to be proved in the operational register that could not be proved in VR-Sets without the formal register. In other words, VR-Forms is a **conservative extension** of VR-Sets.

This secures the freedom of work in the formal register. One may introduce paradoxical terms, terms for uncountable collections, terms for mythological beings — and be confident that the operational register suffers no harm. The boundary between the registers holds formally.

Contents of Part III: formal statement of conservativity (§III.1); main theorem (§III.2); consequences (§III.3); comparison with the consistency of VR-Sets relative to ZF (§III.4); summary (§III.5).

III.1. Statement

Before proving, let us precisely formulate what is being proved.

Languages and theories

Let us denote:

— $\mathbf{L_0}$ — the language of VR-Sets: alphabet and terms without the marker $\ulcorner\cdot\urcorner$.

— $\mathbf{L_1}$ — the language of VR-Forms: alphabet of L_0 plus the marker $\ulcorner\cdot\urcorner$, and with it the possibility of forming a formal term $\ulcorner\tau\urcorner$ from any syntactically correct τ.

— $\mathbf{T_0}$ — the theory of VR-Sets: the closure principle and operational definitions (set, $\varnothing$, $\equiv$, ZFC/ZFA modes).

— $\mathbf{T_1}$ — the theory of VR-Forms: everything that is in T_0, plus the principle of forms (§II.4) and the rules of work with formal terms (§II.5).

Formulas of L_0 are called operational: they speak only about operational sets. Formulas of L_1 may contain formal terms.

Definition III.1 (conservativity)

A theory T_1 is **conservative** over T_0 if for every operational formula $\varphi \in L_0$:

$T_1 \vdash \varphi \iff T_0 \vdash \varphi$.

That is: everything VR-Forms proves about operational sets, VR-Sets could already have proved without the help of the formal register. The formal register adds no new operational force.

What is to be proved

The main theorem of Part III: T_1 is conservative over T_0.

In particular, it follows that if T_0 is consistent, so is T_1.

III.2. Conservativity theorem

Theorem III.1 (conservativity of VR-Forms over VR-Sets)

For every operational formula φ of the language L_0:

$T_1 \vdash \varphi \iff T_0 \vdash \varphi$.

Proof

Direction ($\Longleftarrow$) is trivial: every proof in T_0 is also a proof in T_1 (T_1 contains all axioms and rules of T_0).

Substantive direction ($\Longrightarrow$). Let $T_1 \vdash \varphi$, where $\varphi \in L_0$. We show that $T_0 \vdash \varphi$.

Idea of the proof: every occurrence of a formal term $\ulcorner\tau\urcorner$ in the proof can be **eliminated** by replacement with an equivalent operational formula. More precisely, formal terms are syntactic objects; their use in the proof reduces to manipulations with formulas without operational standing. If the final formula φ is operational, all intermediate formal terms can be «crossed out» by appropriate substitution.

Step 1: translation. Define a mapping π from formulas of L_1 to formulas of L_0:

— For a formula $\psi \in L_0$: $\pi(\psi) = \psi$.

— For an atomic formula $\ulcorner x \in \tau\urcorner$, where τ is the description $\{y : \psi(y)\}$: $\pi(\ulcorner x \in \tau\urcorner) = \psi(x)$ (comprehension, see §II.5).

— For an atomic formula $\ulcorner\tau_1\urcorner = \ulcorner\tau_2\urcorner$, where $\tau_1 = \{y : \psi_1(y)\}$, $\tau_2 = \{y : \psi_2(y)\}$: $\pi(\ulcorner\tau_1\urcorner = \ulcorner\tau_2\urcorner) = \forall y\, (\psi_1(y) \leftrightarrow \psi_2(y))$.

— Connectives and quantifiers — π by induction: $\pi(\neg\alpha) = \neg\pi(\alpha)$, $\pi(\alpha \wedge \beta) = \pi(\alpha) \wedge \pi(\beta)$, $\pi(\forall x\, \alpha) = \forall x\, \pi(\alpha)$, and similarly.

The translation π reduces every formula of L_1 to a formula of L_0: formal terms are replaced by their defining predicates, and the whole formula becomes a statement about properties of objects.

Step 2: preservation of operational formulas. If $\varphi \in L_0$, then $\pi(\varphi) = \varphi$. That is, on operational formulas π is the identity.

Step 3: compatibility of π with the inference rules of T_1. All inference rules of T_1 — rules of classical logic plus axioms of T_0 plus the principle of forms. We show that π translates every inference rule of T_1 into an admissible step in T_0.

(a) **Rules of classical logic** (modus ponens, generalisation, and so on) are compatible with π, since π preserves connectives and quantifiers by definition.

(b) **Axioms of T_0** remain themselves under π — they are already in L_0.

(c) **The principle of forms.** In T_1 the principle of forms states: every description specifies a formal term. Substantively this means that for any description $\{y : \psi(y)\}$ we can form the formal term $\ulcorner\{y : \psi(y)\}\urcorner$ and work with it by the rules of §II.5. Under π this step is translated into work directly with the predicate ψ — that is, into a usual application of the comprehension scheme in L_0, which in T_0 generates nothing new (since the comprehension scheme in T_0 is restricted by the closure principle, see below).

Here is the subtlety. In T_0 comprehension applies to operational descriptions; in T_1 comprehension applies to arbitrary (formal). But π translates the formal term $\ulcorner\{y : \psi(y)\}\urcorner$ back into the predicate ψ. All statements about the formal term become statements about the predicate. And statements about predicates in T_0 are standard first-order statements and do not require operational describability of the predicate.

That is: the formal register serves to say «there is something described by ψ», but π converts this «something» back into simply «property ψ». No new object appears; only a property is there, and one speaks about it in the language of T_0.

Step 4: consequence. If $T_1 \vdash \varphi$, there is a derivation D in T_1 ending in φ. Apply π to each formula of D. We obtain a sequence of formulas of L_0 (by Step 1) in which:

— the first formula π(axiom of T_1) is either an axiom of T_0 (if the original axiom was in T_0) or an admissible application of comprehension (if the original axiom was an instance of the principle of forms).

— every subsequent step π(application of rule) is an admissible step in T_0 (by Step 3).

— the last formula $\pi(\varphi) = \varphi$ (by Step 2).

We obtain a derivation of φ in T_0. Hence $T_0 \vdash \varphi$.

∎

Remark on the method

The method used is a standard technique for conservativity proofs via **interpretation**. The formal register receives an interpretation in the operational: every formal term is a «figure of speech» about a property. Under such an interpretation, everything provable with the use of figures of speech is also provable without them — if the final statement is not itself a figure of speech.

This method is analogous to the interpretation of classes in NBG via formulas of ZFC, or the interpretation of abstractions in λ-calculus via β-reduction. The content is the same: a syntactic extension that does not carry new commitments.

III.3. Consequences

Corollary III.1 (consistency)

If T_0 is consistent, then T_1 is consistent.

Proof. Let T_1 be inconsistent: $T_1 \vdash \bot$. Since $\bot \in L_0$, by Theorem III.1 $T_0 \vdash \bot$. Contradiction. ∎

Since the consistency of T_0 is established in VR-Sets (through the arithmetical equivalence with PA and the consistency of VR-Sets relative to ZF), it follows:

Corollary III.2

VR-Forms is consistent relative to ZF.

Corollary III.3 (safety of paradoxical formal terms)

The introduction into the formal register of the Russell term $\ulcorner\{x : x \notin x\}\urcorner$, the liar, the set of all sets, and other paradoxical descriptions does not generate contradictions in T_1.

Justification. All these terms are syntactically correct descriptions. By the principle of forms they are formal terms. By Theorem III.1 any operational statement derivable in T_1 with their use is derivable also without them. In particular, the contradiction $\bot \in L_0$ is not derivable in T_1 if not derivable in T_0.

Paradoxicality here becomes a **property of the description**, not a source of contradiction. The Russell term $\ulcorner R\urcorner$ is a formal term satisfying the formula $\ulcorner R\urcorner \in \ulcorner R\urcorner \leftrightarrow \ulcorner R\urcorner \notin \ulcorner R\urcorner$. This is a formal statement about a formal term; it shows that $\ulcorner R\urcorner$ **has no operational correlate** (if it had, the contradiction would also be in T_0). And nothing more follows from this.

Corollary III.4 (safety of non-mathematical formal terms)

The introduction into the formal register of ⌜dragon⌝, ⌜god⌝, ⌜soul⌝ and similar descriptions does not generate contradictions in T_1.

Justification. Similar to Corollary III.3. These terms are syntactically correct descriptions; legitimate in the formal register by the principle of forms; their operational correlate is absent; no operational consequences follow from their presence in the formal register.

III.4. Comparison with the consistency of VR-Sets

The chain of consistency in the VR cycle now looks like this:

VR ↔️ **PA** ↪️ **ZF** (established in VR. A Formal System, Part II)

VR-Sets ↪️ **ZF** (established in VR-Sets, through modelling of the operational universe in classical ZF)

VR-Forms ↪️ **VR-Sets** ↪️ **ZF** (established in the present Part III, as conservative extension)

Each link of the chain: consistency of the preceding follows from consistency of the next. If ZF is consistent, all four systems of the VR cycle are consistent.

Remark on strength. Conservativity of VR-Forms over VR-Sets — this is **equal strength** in the operational register, but extended language. VR-Forms can speak of more (including the uncountable, paradoxical, mythological), but proves in the operational register exactly as much as VR-Sets. This is the essence of conservative extension: extending the language without extending the theoretical strength for the original fragment of the language.

Analogy: NBG (Bernays–Gödel) is a conservative extension of ZFC, adding classes as objects. About sets, NBG proves the same as ZFC; but NBG has a more convenient language for speaking about classes. VR-Forms plays the same role relative to VR-Sets: extension of language for talking about the non-operational, without changing operational strength for the operational.

III.5. Summary of Part III

Established:

(1) Distinction of languages and theories of VR-Sets (L_0, T_0) and VR-Forms (L_1, T_1) (§III.1).

(2) Definition of conservativity (Definition III.1, §III.1).

(3) Conservativity theorem: T_1 is conservative over T_0 (Theorem III.1, §III.2). Proof via interpreting mapping π, translating formal terms into their defining predicates.

(4) Consistency of VR-Forms relative to ZF (Corollary III.2, §III.3).

(5) Safety of paradoxical and non-mathematical formal terms (Corollaries III.3, III.4, §III.3).

(6) Place of VR-Forms in the chain of consistency of the VR cycle (§III.4).

What follows. Part IV is devoted to **transits** — rules by which work in the formal register may yield results in the operational register. The conservativity theorem guarantees the safety of transits; Part IV provides their positive content: when and how the formal register is able to **help** operational proofs.

Part IV. Transits

IV.0. Introduction

Part III established the conservativity of VR-Forms over VR-Sets. This is a negative result: the formal register **does not generate** new operational statements. Part IV provides the positive content: the formal register **helps** in reasoning about operational sets, even without adding new theorems.

The help consists in the following. Many classical proofs use uncountable intermediaries: classical $\mathbb{R}$, σ-algebras, Lebesgue measure, the axiom of choice for uncountable families. These intermediaries are not themselves operational. But if a proof using them ends in an operational statement, then — by conservativity — this statement is provable also without the intermediaries. Part IV formulates the **transit**: a rule by which a formal proof of an operational statement is recognised as a proof in VR-Sets.

Contents: notion of transit (§IV.1); transit rule and its justification (§IV.2); notion of witness (§IV.3); safe and problematic transits (§IV.4); examples (§IV.5); summary (§IV.6).

IV.1. The notion of transit

A transit is the passage from the formal register to the operational. More precisely: the use of formal terms as intermediate means in a proof whose result is an operational statement.

Definition IV.1 (transit)

A **transit** is a derivation in T_1 in which:

(a) the final formula φ is operational ($\varphi \in L_0$);

(b) in intermediate steps of the derivation formal terms are used (a nonzero number of steps contains the marker $\ulcorner\cdot\urcorner$).

A transit, then, is a derivation that **passes through** the formal register on the way to an operational result.

What conservativity promises

By Theorem III.1, if such a derivation exists in T_1, there also exists a derivation of φ in T_0 (without formal terms). That is, every transit **can be rewritten** as an operational proof.

However, the rewriting may require substantial work. In practice, a formal derivation can be shorter and clearer than the equivalent operational one. A transit is a way of **using** the formal register to ease the work, knowing in advance that the result is operational.

IV.2. The transit rule

Let us formalise how exactly to use a transit.

Transit rule

If a derivation D ending in an operational formula φ is constructed in T_1, then φ is accepted as proved in T_0.

Justification. By Theorem III.1. The existence of a derivation in T_1 of an operational formula is equivalent to the existence of a derivation in T_0.

The transit rule, then, is not a new inference rule. It is a **methodological permission**: one may work in the formal register, and the result automatically belongs to the operational, provided it is itself operational.

What transit does not do

A transit does not turn formal terms into operational objects. The Russell term ⌜R⌝, used in a transit, does not become a set. It remains a formal term used as an intermediate syntactic element. Only the final formula, if it is operational, passes into the operational register.

A transit does not allow anything to be proved in T_0 that could not be proved without the transit. It is only a **convenient way** to conduct a proof. No new strength is provided by the transit.

A transit does not «justify» formal terms. That a certain formal term is used in a transit does not mean it is operationally realisable (see §II.7). Many formal terms figure in transits while remaining non-realisable. Realisability is a separate question; transit is a syntactic procedure.

IV.3. Witness

Connected with transit is the notion of **witness** — an operational object whose existence is proved by the transit.

Definition IV.2 (witness)

Let a formula φ have the form ∃x ψ(x), where ψ is operational. A **witness** of φ is an operational set A such that ψ(A) is true in T_0.

A transit may prove ∃x ψ(x) — that is, the existence of a witness — without exhibiting a specific A. This is the typical situation of non-constructive proofs in classical mathematics.

Extraction of a witness

If a transit yields $\exists x\, \psi(x)$, then by conservativity there is a derivation of $\exists x\, \psi(x)$ in T_0. However, this derivation may not be constructive: it establishes existence without exhibiting x.

The existence of a **witness extraction** is a separate question. For constructive proofs existence entails construction; for classical it does not necessarily. VR-Forms inherits this distinction: transit yields an operational statement but does not always yield an operational construction.

Remark

This is the place where the boundary between two possible roles of VR-Forms is visible.

Weak role (proof of existence): transit is used to establish that an operational object with a given property exists. The object itself may remain unspecified.

Strong role (construction): transit is used as a heuristic — formal reasoning suggests how to build an operational object. The construction is then verified in the operational register.

Both roles are admissible. In practice the strong role is preferable, since it yields an explicit operational object, not leaving it «known only formally».

IV.4. Safe and problematic transits

All transits are safe in the sense of consistency (Part III). However, from a practical viewpoint **direct** and **problematic** transits can be distinguished.

Direct transits

A transit is called **direct** if in it formal terms are used as convenient abbreviations for operational objects. That is, all intermediate formal terms are operationally realisable (§II.7), and the use of the formal register is mere syntactic convenience.

Direct transits trivially rewrite into operational proofs. Their use is a stylistic decision.

Problematic transits

A transit is called **problematic** if it essentially uses **non-realisable** formal terms. That is, intermediate steps refer to ⌜$\mathbb{R}$⌝, ⌜$\wp(\mathbb{R})$⌝, ⌜the Vitali set⌝, to choice functions over uncountable families, and so on.

Here one must be more careful. Conservativity guarantees that the result of a problematic transit is translatable into the operational register — but the rewriting may be cumbersome, and sometimes require substantial operational work.

A principal observation

Problematic transits are places where classical mathematics essentially leans on uncountability. If the result of a transit is operational (for example, a statement about a concrete computable function), then by conservativity it is provable also without uncountable intermediaries. **This is the programme of VR-Audit** — analysis of classical theorems with the aim of extracting the operational remainder.

Within Part IV: established only that the rewriting is possible. **How** to conduct it in concrete cases — the subject of Parts V onward, and in full — the programme of VR-Audit.

IV.5. Examples

Example 1: proof of the countability of the set of algebraic numbers

Classical proof: the set of polynomials with rational coefficients is countable (as a countable union of countable sets); each polynomial has finitely many roots; therefore there are countably many algebraic numbers.

This proof is already operational: all its steps concern operational sets ($\mathbb{Q}$, polynomials with rational coefficients, roots). No transit is needed. The result belongs to T_0.

Example 2: proof of Bessel's inequality

Bessel's inequality in Hilbert space: for any orthonormal family $\{e_n\}$ and any x in the space, $\Sigma|\langle x, e_n\rangle|^2 \leq \|x\|^2$.

The classical proof uses the formulation «Hilbert space», which in the full classical form is uncountable. However, the inequality itself is a statement about operational objects, if x and $\{e_n\}$ are computable. Transit: using ⌜Hilbert space⌝ as a formal term in intermediate steps, we obtain an operational result — the inequality for concrete computable x and $\{e_n\}$.

By conservativity this means that Bessel's inequality for computable data is provable in VR-Sets directly. The transit here is convenience, not necessity.

Example 3: a proof essentially using the uncountable

The Hahn–Banach theorem in its general classical formulation: every bounded linear functional given on a subspace extends to the whole space with preservation of norm.

The classical proof uses Zorn's lemma, which depends on AC for uncountable families. The formal term ⌜functional on the whole $\mathbb{R}$-space⌝ is essentially non-realisable.

Transit: if the result of the theorem is a statement about a computable extension of a computable functional, then by conservativity the result is provable in VR-Sets. But the rewriting here is non-trivial: an explicit operational construction of the extension is needed, which in the general case requires additional work. This is an example of a **problematic transit**, requiring operational labour to extract the operational content.

Example 4: spectral theorem for the harmonic oscillator

The harmonic oscillator has discrete spectrum $E_n = \omega(n + 1/2)$. The classical formulation of the spectral theorem uses ⌜projection-valued measure⌝ — a formal term requiring an uncountable σ-algebra.

But for the harmonic oscillator this formal apparatus is superfluous: the discrete spectrum and eigenfunctions (Hermite polynomials multiplied by a Gaussian) are operational. The transit here yields an operational statement — for example, computability of matrix elements of the energy operator — without essential dependence on uncountable structure.

This is an example where the formal register is **replaced** by a direct operational construction. The transit here is a methodological permission to use the classical proof, but more valuable content is provided by the direct operational derivation, which we build directly in T_0.

IV.6. Summary of Part IV

Established:

(1) Transit as a derivation in T_1 passing through formal terms and ending in an operational formula (Definition IV.1, §IV.1).

(2) Transit rule: every transit yields a proof in T_0 (§IV.2). Justification — Theorem III.1.

(3) The notion of witness and the distinction of the weak (existence) and the strong (construction) roles of transit (§IV.3).

(4) Distinction of direct and problematic transits (§IV.4); problematic — places where classical mathematics essentially leans on the uncountable.

(5) Examples: algebraic numbers (no transit), Bessel's inequality (direct transit), Hahn–Banach (problematic), harmonic oscillator (direct operational replacement) (§IV.5).

Methodological role of Part IV. Part IV closes the technical apparatus of VR-Forms. Onward we pass to substantive **application** of the apparatus to classical mathematical (Part V) and non-mathematical (Part VI) formal terms.

Connection with VR-Audit. Part IV outlines a programme: for each classical theorem φ, determine whether its proof is a direct transit (i.e. φ is provable in VR-Sets without essential loss), a problematic transit (requires operational labour to extract content), or not a transit at all (φ is non-operational, remains in the formal register). Systematic conduct of such an analysis is the content of the future work VR-Audit; VR-Forms provides for it the apparatus.

Part V. Forms of Mathematics

V.0. Introduction

Parts II–IV gave the apparatus of VR-Forms. Part V applies it to classical mathematics. The aim is to demonstrate how uncountable, paradoxical and non-operational mathematical objects find their place in the formal register, and how this changes their status.

Method of exposition: for each characteristic class of formal terms we give (1) the classical description, (2) status in the VR-Sets operational register, (3) status in the formal register of VR-Forms, (4) methodological consequences.

Contents of Part V: classical ℝ as formal term (§V.1); classical ℘(ℕ) (§V.2); the Vitali set and AC for uncountable families (§V.3); the Russell term and paradoxical classes (§V.4); proper classes (NBG) and their status (§V.5); summary (§V.6).

V.1. Classical ℝ

Classical description

Classical ℝ — the set of all real numbers, defined in one of equivalent ways: as the complete ordered field, as the set of Dedekind cuts of ℚ, as the set of equivalence classes of fundamental sequences in ℚ. Cardinality $2^{\aleph_0}$ — uncountable.

Status in VR-Sets

In VR-Sets there is ℝ_VR — the operational set of computable reals, isomorphic to the field of computable reals (VR-Numbers §V.4). ℝ_VR is countable: each computable real is given by an algorithm, algorithms are countable.

The full classical ℝ is operationally absent in VR-Sets. Not as «forbidden», but as a description signifying nothing: «the set of all reals» operationally does not specify a functionality (there is no procedure that would enumerate all reals).

Status in VR-Forms

⌜ℝ⌝ is a formal term. Admissible in the formal register by the principle of forms. Not operationally realisable.

In the formal register one can formulate statements about ⌜ℝ⌝: «the cardinality of ⌜ℝ⌝ is $2^{\aleph_0}$», «⌜ℝ⌝ is complete as a metric space», «⌜ℝ⌝ is ordered». All these statements belong to the formal register. They do not generate operational objects and have no operational force.

Methodological consequences

Statements of the form «property P holds for all reals» in the formal register speak of $\ulcorner\mathbb{R}\urcorner$. To translate such a statement into the operational register, one must show that for all elements of $\mathbb{R}_{VR}$ (i.e. for all computable reals) the property P holds. This is passage via transit (§IV.2).

If P holds for all $\mathbb{R}_{VR}$ — we obtain an operational statement. If P holds for the classical $\ulcorner\mathbb{R}\urcorner$ but is not derivable for $\mathbb{R}_{VR}$ without essential dependence on uncountability — this is the case of a **problematic transit** (§IV.4).

In practical mathematics most statements about $\mathbb{R}$ relate to properties derivable from computability (completeness, ordering, field, metric). These statements transit directly. Statements essentially using uncountability (Liouville's theorem on transcendental numbers, the existence of measurable non-recursive sets) require separate analysis.

V.2. Classical $\wp(\mathbb{N})$

Classical description

$\wp(\mathbb{N})$ — the set of all subsets of $\mathbb{N}$. By Cantor's theorem it has cardinality $2^{\aleph_0}$, uncountable.

Status in VR-Sets

$\wp_{VR}(\mathbb{N})$ — the operational set of describable subsets of $\mathbb{N}$ (VR-Sets §III.5). Countable. Contains all computable subsets of $\mathbb{N}$.

The statement «$\wp_{VR}(\mathbb{N})$ is countable» is true in the operational register. Cantor's diagonal argument operationally does not work for the same reasons as in Computable Analysis: there is no effective enumeration of all describable subsets (this is equivalent to the halting problem), so the diagonal construction is not operational.

Status in VR-Forms

$\ulcorner\wp(\mathbb{N})\urcorner$ is a formal term. In the formal register the statement «$\ulcorner\wp(\mathbb{N})\urcorner$ has cardinality $2^{\aleph_0}$» is admissible. The statement «$\ulcorner\wp(\mathbb{N})\urcorner$ is uncountable» is also admissible. Both belong to the formal register.

Here the difference of the two registers becomes visible. In the operational: $\wp_{VR}(\mathbb{N})$ is countable. In the formal: $\ulcorner\wp(\mathbb{N})\urcorner$ is uncountable. No contradiction — these are statements about **different** objects. $\wp_{VR}(\mathbb{N})$ is an operational set; $\ulcorner\wp(\mathbb{N})\urcorner$ is a formal term; they are not identical.

Skolem's paradox in VR-Forms

In the classical context Skolem's paradox is the observation that ZFC has a countable model in which «$\wp(\mathbb{N})$ is uncountable» is true (Löwenheim–Skolem). This is regarded as disconcerting: «how can $\wp(\mathbb{N})$ be countable from outside but uncountable from inside?»

In VR-Forms Skolem's paradox is removed explicitly. The «inside/outside» distinction becomes a distinction of registers. $\wp$_VR(ℕ) is countable operationally («outside», in the full picture). ⌜$\wp$(ℕ)⌝ is uncountable formally («inside», as a statement about a formal term). These are two different objects, each with its true status in its own register.

V.3. The Vitali set and AC for uncountable families

Classical description

The Vitali set — a choice of one representative from each equivalence class on ℝ under the relation $x - y \in \mathbb{Q}$. The existence requires AC for an uncountable family (the number of equivalence classes is uncountable). Not Lebesgue measurable.

Status in VR-Sets

The Vitali set is operationally absent in VR-Sets. The description «choose one from each class» is not operational: there is no procedure that would, on the request «give the next class», yield a definite representative (classes are uncountable). AC in VR-Sets is a theorem about countable families (VR-Sets §III.9 and §VI.4), not applicable to uncountable.

The Banach–Tarski paradox is absent in VR-Sets by construction, since it uses precisely Vitali-like uncountable choices (VR-Sets §VI.5).

Status in VR-Forms

⌜V⌝ is the formal term for the Vitali set. In the formal register the statement «⌜V⌝ exists» is admissible (by the principle of forms every description specifies a formal term). The statement «⌜V⌝ is non-measurable» — formal.

⌜Banach–Tarski⌝ — a formal theorem about a paradoxical decomposition of ⌜the ball⌝ into a finite number of parts and assembly of two ⌜balls⌝. In the formal register admissible as a formal theorem. In the operational register no analogue.

Methodological consequences

If in some context Vitali is used (for example, to construct a non-measurable set), this can be described in the formal register, but the operational correlate is absent. All applications essentially requiring Vitali remain in the formal register.

On the other hand, many classical results formally using AC for uncountable families can be rewritten without it — this is the programme of Reverse Mathematics. VR-Forms provides for this programme a natural frame: check whether the use of uncountable AC is essential (i.e. the transit is problematic in the sense of §IV.4) or eliminable (direct transit).

V.4. The Russell term and paradoxical classes

Classical description

Russell's class $R = \{x : x \notin x\}$. In naive set theory — source of the paradox: $R \in R \leftrightarrow R \notin R$. In ZF dissolved by the axiom of separation: one can form only $\{x \in A : x \notin x\}$ for an already existing A, and this does not generate R as a whole.

Status in VR-Sets

R is not an operational set (VR-Sets §II.3). The description «a functionality that yields x if and only if x does not yield itself» contains a contradiction on the query «$R \in R$?». The closure principle does not apply to contradictory descriptions.

Status in VR-Forms

$\ulcorner R \urcorner$ is a legitimate formal term. The principle of forms does not forbid contradictory descriptions (§II.4): syntactic correctness suffices.

In the formal register the statement $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ is admissible. This is a formal statement about a formal term, derivable directly from the definition of $\ulcorner R \urcorner$.

This generates no contradiction in T_1. Proof: if from $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ one could derive $\perp \in L_0$ (an operational contradiction), then by conservativity $\perp$ would be derivable also in T_0, which is not the case.

A new reading of Russell

In the classical setting Russell's paradox is an indication of the necessity to restrict the formation of sets. In VR-Forms Russell's paradox has a different meaning: it shows that the description $\ulcorner R \urcorner$ **has no operational correlate**. This is not a defect of the description, but a property of it: some formal terms are operationally non-realisable.

Paradoxicality is a property of form, not an operational failure. The liar, Berry, Grelling, Burali-Forti — all classical paradoxes in VR-Forms become statements about properties of formal terms, saying that the corresponding descriptions have no operational correlate.

Remark on the freedom of the formal register

The admissibility of contradictory formal terms might seem alarming — but it does not break classical logic in the formal register. Logical operations apply to formulas as usual, and from a local contradiction (such as $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$) no global contradiction follows.

The point is that the contradiction $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ is not a statement «is and is not at once»; it is a statement «the formal term $\ulcorner R \urcorner$ has such a property». The property characterises $\ulcorner R \urcorner$, separating it from operational sets. No universal logical catastrophe follows from it.

V.5. Proper classes

Classical description

In NBG (Bernays–Gödel) and MK (Morse–Kelley) extensions of ZFC, **proper classes** are introduced — collections too «large» to be sets. The class of all sets V, ordinals Ord, cardinals Card — proper classes. They are not elements of any set.

Status in VR-Sets

In VR-Sets proper classes are absent in the classical form, since the whole universe is countable (VR-Sets §VI.1) and the «too large» in the classical sense does not arise. However, there is an analogue: **describable families of functionalities that are not themselves functionalities**. For instance, «the family of all computable functions ℕ → ℕ» is a describable family but not an operational set (no procedure for their effective enumeration; this reduces to the halting problem).

Status in VR-Forms

⌜V⌝ (the class of all sets) is a formal term. Admissible in the formal register. ⌜Ord⌝, ⌜Card⌝ — also formal terms.

The formal register plays here the same role as classes in NBG: it allows one to speak about large collections without introducing them into the operational universe. Difference: in NBG classes are second-sort objects; in VR-Forms formal terms are formulas, not objects.

The hierarchy of t-applications

In VR-Sets §VII.3 the hierarchy of applications of the successor t was discussed as an operational analogue of Tarski's hierarchy of metalanguages. In VR-Forms this hierarchy receives an additional reading.

Each t^n(ω) is an operational set. Their union, «the entire hierarchy t^n», is a describable family that is not an operational set (it is an unbounded process). In the formal register it is a formal term ⌜⋃_n t^n(ω)⌝ — an operational analogue of a proper class. This is an illustration: in VR-Forms the «too large» does not disappear; it passes into the formal register, remaining workable as a figure of speech.

V.6. Summary of Part V

Established:

(1) Classical ℝ is a formal term ⌜ℝ⌝, not identical with the operational ℝ_VR (§V.1).

(2) Classical ℘(ℕ) is a formal term ⌜℘(ℕ)⌝, not identical with the operational ℘_VR(ℕ); Skolem's paradox is removed as a difference of registers (§V.2).

(3) The Vitali set and the Banach–Tarski paradox are formal terms; in the operational register they are absent (§V.3).

(4) The Russell class is a legitimate formal term with a contradictory characteristic; paradoxicality is reinterpreted as a property of form indicating operational non-realisability (§V.4).

(5) Proper classes of NBG correspond to formal terms of VR-Forms; the hierarchy of t-applications provides the operational analogue of a proper class (§V.5).

General conclusion. The formal register of VR-Forms covers the whole classical set theory — uncountable collections, paradoxical classes, large classes of NBG — without placing any burden beyond the countable operational universe. Classical mathematics becomes **speech** about formal terms, operationally realisable only in part; the programme of VR-Audit consists in a systematic classification of what in this speech has an operational correlate and what does not.

What follows. Part VI extends the formal apparatus to non-mathematical descriptions — dragons, gods, philosophical categories — demonstrating the universality of the formal register. This is a natural completion of applications: the formal language of VR-Forms works with speech without exception, not only mathematical.

Part VI. Forms of Speech

VI.0. Introduction

Part V applied the formal apparatus to classical mathematics: classical $\mathbb{R}$, $\wp(\mathbb{N})$, Vitali, Russell's class, proper classes — all received the status of formal terms. Part VI extends the application beyond mathematics: to mythological beings, theological concepts, literary figures, philosophical categories.

This is not a rhetorical move and not an artistic extension. It is a **test of the universality** of the formal register. If the formal register really covers speech without exception, it must work uniformly with mathematical and non-mathematical descriptions. Part VI demonstrates exactly this.

Contents: universality of the formal register (§VI.1); mythological terms (§VI.2); theological terms (§VI.3); philosophical categories as formal terms (§VI.4); literary figures (§VI.5); status of non-mathematical terms and its philosophical significance (§VI.6); summary (§VI.7).

VI.1. Universality of the formal register

The principle of forms (§II.4) states: every syntactically correct record of a description specifies a formal term. No restrictions on the content of the description are imposed. The description may be mathematical, may be non-mathematical, may refer to the existing, the non-existing, the imagined, the paradoxical. The principle of forms applies to all uniformly.

This means: the formal register of VR-Forms is a **universal language of descriptions**. Everything for which a correct syntactic description can be formed has a place in it as a formal term.

Remark on syntactic correctness

«Syntactic correctness» in VR-Forms is a formal notion, set by the rules of §II.1. A description is correct if it is built by these rules. This means the syntax of VR-Forms is a fixed formal language; descriptions in natural language («a fire-breathing dragon», «an omnipotent god») are not directly expressed in it.

However, every meaningful natural-language description can be translated into the formal — if a definite predicate is assigned to it. «Dragon» becomes a formal term ⌜{x : x is a fire-breathing lizard-like creature with wings}⌝, where «is a fire-breathing lizard-like creature with wings» is a predicate expressed in the language of VR-Forms (via a chosen system of feature designations). Similarly with «god», «soul», «unicorn».

Thus the non-mathematical terms in VR-Forms are not literary figures as such, but **their formalisations** through predicates. The choice of formalisation depends on context;

different contexts may give different «dragons» (Chinese, European, mythological, literary), and each formalisation is its own formal term.

VI.2. Mythological terms

Dragon

⌜dragon⌝ is a formal term. Formalisation: ⌜$\{x : \text{fire\_breathing}(x) \wedge \text{lizard\_like}(x) \wedge \text{winged}(x)\}$⌝, where the predicates «fire_breathing», «lizard_like», «winged» are elements of the chosen feature system.

In the operational register of VR-Sets: the corresponding operational set is absent. The description «a functionality enumerating all fire-breathing lizard-like winged creatures» is not a procedure over $\varnothing$; no operational action realises it.

In the formal register: ⌜dragon⌝ is a formal term. One can speak of it, formulate statements: «⌜dragon⌝ breathes fire» (true in the formal register by definition of ⌜dragon⌝), «⌜dragon⌝ exists» — a statement of the formal register, having no operational force.

Unicorn, phoenix, centaur, minotaur

Analogously. Each is a formal term with its own formalisation:

⌜unicorn⌝ = ⌜$\{x : \text{horse\_like}(x) \wedge \text{single\_horned}(x)\}$⌝

⌜phoenix⌝ = ⌜$\{x : \text{bird\_like}(x) \wedge \text{reborn\_from\_ashes}(x)\}$⌝

⌜centaur⌝ = ⌜$\{x : \text{has\_human\_upper\_body}(x) \wedge \text{has\_horse\_lower\_body}(x)\}$⌝

All formal terms. None operationally realisable. For each, one can formulate correct statements in the formal register that follow from the definition.

Remark on biological consistency

Some mythological terms are formally **contradictory** in a strong sense — that is, their definitions rely on properties biologically incompatible. For example, ⌜centaur⌝ with two cardiovascular systems, or ⌜dragon⌝ with an endogenous fire organ without thermal damage.

This does not make the formal term inadmissible in VR-Forms. The principle of forms does not require biological or physical consistency. Syntactic correctness suffices.

This is an analogue of the mathematical case in which contradictory formal terms are admissible (Russell's class). In the formal register contradictoriness is a property of the description, not a prohibition.

VI.3. Theological terms

God in monotheistic understanding

⌜god⌝ = ⌜{x : unique(x) ∧ omnipotent(x) ∧ eternal(x) ∧ creator_of_world(x)}⌝ — a formal term. The formalisation is the standard theological definition translated into predicate form.

In the operational register of VR-Sets: the corresponding operational set is absent. No procedure over ∅ realises «the unique omnipotent eternal creator». Not as a «prohibition» but as a description signifying nothing operationally: operationally not reproducible.

In the formal register: ⌜god⌝ is a formal term. One can speak of it, prove statements following from the definition: ⌜god⌝ is unique (by definition), ⌜god⌝ created ⌜the world⌝ (if ⌜the world⌝ is a separate formal term), and so on.

Classical theological paradoxes

The paradox of the stone (can an omnipotent god create a stone he cannot lift?) in VR-Forms is treated as a statement of the formal register about the properties of ⌜god⌝, in particular about the consistency of the predicate «omnipotent». If the formalisation of omnipotence is contradictory (as in this paradox), then ⌜god⌝ in the given formalisation contains a contradictory characteristic — analogously to the Russell term. This is a property of the **description**, not an operational failure.

This reinterpretation is consistent with the general position of VR-Forms: paradoxicality of theological concepts is a property of the chosen formalisation. Other formalisations (omnipotence as «can do all that is logically possible») generate no such contradiction; and these other formalisations are their own formal terms.

God in polytheistic and other systems

⌜Zeus⌝, ⌜Odin⌝, ⌜Brahma⌝, ⌜Allah⌝ — each a formal term with its own formalisation. Each description specifies its own formal term. Statements about them belong to the formal register.

VR-Forms neither asserts nor denies the reality of these entities. It only states: in the operational universe of VR they are not (for the same reasons as unicorns: no operational action over ∅). In the formal register they are admissible as formal terms. About their existence **outside** the operational universe of VR the present work does not speak.

VI.4. Philosophical categories

Classical philosophical categories — universals, essence, accident, form, matter, soul, mind — are formal terms in VR-Forms by the same principle: each description, regardless of its philosophical tradition, satisfies syntactic correctness and so specifies a formal term.

Universals

«Universal» in the classical philosophical tradition — a common nature shared by many individuals. ⌜universal⌝ is a formal term. Each concrete instance (⌜redness⌝, ⌜humanity⌝, ⌜triangularity⌝) is its own formal term with its own formalisation.

The medieval dispute of realists and nominalists is a dispute about whether universals have operational correlates. In terms of VR-Forms: whether formal terms like ⌜redness⌝ have an operational realisation.

The position of VR-Forms is closer to nominalism: in VR — where there are only operations — universals as self-standing objects are absent. They are admissible as formal terms — that is, as names, not entities. This is a precise formulation of medieval nominalism in the modern formal apparatus.

Aristotelian and scholastic categories

⌜essence⌝, ⌜accident⌝, ⌜form⌝, ⌜matter⌝ — all formal terms, with formalisations depending on the chosen scholastic system (Aristotle, Aquinas, Suárez each give different formalisations). VR-Forms does not adjudicate classical metaphysical disputes; it states only that these categories are formal terms, work with them belongs to the formal register, and their operational realisability in VR-Sets is absent.

Soul, spirit, mind

⌜soul⌝, ⌜spirit⌝, ⌜mind⌝ — formal terms. Whether ⌜mind⌝ can be reduced to the operational (for instance, to a procedure over a physically realised neural network) is a separate question — about the existence of an operational realisation of the corresponding formal term. VR-Forms does not answer it; it provides a **language** for formulating it.

VI.5. Literary figures

Literary characters and worlds form a particularly rich domain of formal terms.

Characters and worlds

Literary characters (⌜Hamlet⌝, ⌜Sherlock Holmes⌝, ⌜Anna Karenina⌝) are formal terms. Each carries an extensive formalisation, set by the corresponding literary work.

Literary worlds (⌜Middle-earth⌝, ⌜Hogwarts⌝, ⌜Narnia⌝) are also formal terms, often containing their own formal terms (⌜elf⌝, ⌜hobbit⌝, ⌜Harry's wand⌝) — nested structures of formal terms.

Truth in literature

The statement «Sherlock Holmes lived on Baker Street» — true in what sense? In VR-Forms this is a formal-register statement about the formal term ⌜Sherlock Holmes⌝. Truth is determined by the chosen formalisation (the text of the work as a source of predicates).

«Sherlock Holmes exists» in VR-Forms: in the formal register admissible (⌜Sherlock Holmes⌝ is a formal term, in this sense «exists»); in the operational register the operational correlate is absent.

This two-register treatment corresponds to the intuitive distinction «exists in the work» vs «exists in reality», but gives it a formal form.

VI.6. Status of non-mathematical terms: philosophical significance

Let us list what has been shown in Part VI, and what conclusions follow.

(1) The formal register is universal

Demonstrated: the formal apparatus of VR-Forms works with mathematical (Part V) and non-mathematical (the present part) descriptions uniformly. The principle of forms and the transit rule have no restriction by subject matter.

This asserts the universal character of the two-register structure: the division into operational and formal applies to speech in general, not specifically to mathematics.

(2) Status of non-mathematical terms

In VR — where there are only operations — non-mathematical formal terms (dragons, gods, unicorns) have no operational correlate. This is not a denial of their reality. It is a statement: VR does not realise them. Whether they exist in another ontology (theological, literary, psychological) is a separate question, not pertaining to VR-Forms.

The distinction «not in the operational universe of VR» and «not at all» is principial. VR-Forms does not claim exclusivity for its operational universe. It only clearly fixes it, and relative to it classifies formal terms.

(3) Connection with nominalism

VR-Forms gives a precise formal form to the classical nominalist position: universals and general entities are names (formal terms), only the concrete (the operational) is real. Difference from classical nominalism: VR-Forms replaces «the concrete» with «the operational» (what in VR has an operational correlate).

This is a consistent extension of nominalism from objects to operations: real are not «concrete objects» (which in modern physics is already questionable) but operational actions upon the fundamental $\varnothing$. Names — everything else, including «humanity», «redness», «Socrates as a man» in the universal sense.

(4) Substantive role of the formal register in non-mathematical speech

What practical significance does the formal register have for non-mathematical descriptions?

Above all, **methodological**. Statements about ⌜dragon⌝, ⌜god⌝, ⌜soul⌝ can be formulated and discussed formally, without claiming their reality. This removes a classical difficulty: «is there sense in talking about the non-existent?» — answered by: the formal register provides meaningful speech without operational commitments.

This is relevant to theology (one may build theological systems as formal structures without making claims about their reality), literary studies (literary worlds as formal structures), philosophy (categories as formal terms), mythology (mythological systems as formal systems).

VR-Forms does not take away from these areas their content. It provides them with an **apparatus** for talking about them, clearly separating the internal logic of the system from the question of its reality.

VI.7. Summary of Part VI

Established:

(1) The formal register of VR-Forms is universal: it covers mathematical and non-mathematical descriptions uniformly (§VI.1).

(2) Mythological terms (dragons, unicorns, phoenixes) are formal terms, not operationally realisable in VR (§VI.2).

(3) Theological terms (god, divine entities) are formal terms; classical theological paradoxes are treated as properties of the formalisation (§VI.3).

(4) Philosophical categories (universals, essence, soul) are formal terms; VR-Forms provides a formal form for classical nominalism (§VI.4).

(5) Literary figures and worlds are formal terms; «truth in the work» receives a precise formal reading (§VI.5).

(6) Philosophical significance of the universality of the formal register: clarity about what VR realises, without a claim of exclusivity; methodological usefulness for theology, literary studies, philosophy (§VI.6).

Substantive summary. VR-Forms is not only a formal apparatus for mathematical foundations. It is a **universal language of speech about the non-operational**. Dragons and classical $\mathbb{R}$ have in it the same status: formal terms, legitimate in the formal register, having no operational correlate in the operational universe of VR. This is the position that in Part I was stated verbally; the present part demonstrates it in operation.

What follows. Part VII is devoted to the logic of the two-register system — what inference rules act in each register, how they interact, whether classical logic is preserved. This is a technical closure of the system.

Part VII. Two-Register Logic

VII.0. Introduction

The apparatus of VR-Forms was built in Parts II–IV; applications were given in Parts V–VI. The present part fixes the **logical properties** of the two-register system: which rules of inference act, how the two registers interact, whether classical logic is preserved, what peculiarities arise on the boundary.

Part VII is the technical closure of the system. No new operational commitments are accepted here. The aim is to collect in one place the logical properties scattered through the preceding parts and to make their consequences visible.

Contents: classical logic in both registers (§VII.1); mixed formulas and rules of work with them (§VII.2); quantifiers in the formal register (§VII.3); the law of the excluded middle and its status (§VII.4); connection with two-sorted logic and other formal systems (§VII.5); summary (§VII.6).

VII.1. Classical logic in both registers

The principled decision, fixed in Part II §II.5 and justified by Theorem III.1: VR-Forms uses **classical first-order logic** in both registers.

What this means

In the operational register the usual rules of classical logic act: modus ponens, generalisation, axiom schemes of classical predicate calculus. This is inherited from VR-Sets — there too classical logic (VR-Sets §I.3).

In the formal register — the same rules. The formal register is not a new logic. It is the same logical apparatus applied to a new type of objects (formal terms).

Why classical, not intuitionistic

A common solution in the foundations of mathematics for operational systems is to pass to intuitionistic logic (Brouwer, Bishop). VR-Sets has already declined this move: classical logic is preserved (see VR-Sets §VIII.3, comparison with constructivism).

The reason in VR-Sets: operational definiteness is already built into the closure principle. Every operational functionality gives a definite answer to every query; «a third» among operations is absent. So the law of excluded middle holds not as a logical axiom but as a property of operationality.

In VR-Forms the situation is different: in the formal register there may be contradictory descriptions (Russell's class, the liar paradox). Here classical logic holds for another reason — because formal terms are formulas, and working with them is working with formulas; standard logic applies to formulas as usual. Contradictions in formal terms do not generate global contradictions, since the formal register is conservative over the operational (Part III).

Remark on another possibility

VR-Forms could be considered with intuitionistic or paraconsistent logic in the formal register. That would lead to a different system. The present work chooses classical logic for simplicity and for compatibility with classical mathematics during transits (Part IV). Alternative variants remain open directions.

VII.2. Mixed formulas

Formulas of VR-Forms may simultaneously contain operational terms and formal ones. For instance: «every element of A satisfies ψ», where A is an operational set and ψ is a predicate referring to a formal term. Or: «$\ulcorner\tau\urcorner$ has an element in A», where $\ulcorner\tau\urcorner$ is formal and A is operational.

Rules for working with mixed formulas

Mixed formulas are admissible in L_1 by the principle of forms and the usual syntactic rules. Logical operations apply to them standardly.

In conducting a derivation, a mixed formula may be transformed either into a purely operational one (if all formal terms are eliminable — for example, are operationally realisable), or remain mixed. There is no requirement to «squeeze out» formal terms — they may remain in the derivation as intermediate elements.

Translation π on mixed formulas

The mapping π from §III.2 applies to mixed formulas naturally: operational terms remain unchanged, formal terms are translated into their defining predicates.

In particular, the formula $\ulcorner x \in \tau\urcorner \wedge x \in A$, where $\tau = \{y : \psi(y)\}$ and A is operational, translates into $\psi(x) \wedge x \in A$. We obtain a purely operational formula. This is the mechanism of conservativity (§III.2).

Substantive significance

Mixed formulas are the principal working tool in the practice of VR-Forms. They allow one to formulate statements of the form «the operational object A satisfies the property described formally». This is the typical situation of transits: the formal register provides a convenient language for the property, the operational provides a concrete object.

Example: «$\wp$_VR(ω) is a subfamily of $\ulcorner\wp(\omega)\urcorner$» — a mixed formula stating that the operational set of describable subsets of ω is a part of the formal term «classical $\wp(\omega)$». The statement is admissible in the formal register; in the operational register it is equivalent to «every element of $\wp$_VR(ω) satisfies the predicate 'is a subset of ω'», which is trivially true.

VII.3. Quantifiers in the formal register

Quantifiers ∀ and ∃ applied to formal terms require a separate comment — they have specific content.

Quantifier ∃ over formal terms

The notation $\exists\ulcorner\tau\urcorner\,\varphi(\ulcorner\tau\urcorner)$ means: «there exists a formal term $\ulcorner\tau\urcorner$ such that φ». Since formal terms are syntactic records, this is equivalent to: «there exists a syntactically correct record of a description τ for which the formula $\varphi(\ulcorner\tau\urcorner)$ is true».

The set of all formal terms is countable (by the finiteness of the alphabet and the inductive definition of terms). Hence ∃-quantification over formal terms is, in essence, quantification over a countable set of syntactic objects. This does not generate uncountable collections, does not require choice over uncountable families, does not violate the countability of the universe of VR-Sets.

Quantifier ∀ over formal terms

The notation $\forall\ulcorner\tau\urcorner\,\varphi(\ulcorner\tau\urcorner)$ means: «for every formal term $\ulcorner\tau\urcorner$, φ holds». Analogously: quantification over a countable set of syntactic objects.

Remark: in this quantification participate **all** formal terms — including non-realisable, contradictory, mythological. A statement $\forall\ulcorner\tau\urcorner\,\varphi(\ulcorner\tau\urcorner)$ is a statement about all descriptions without exception. This is a strong statement, and in practice rarely encountered; more often the bounded quantification $\forall\ulcorner\tau\urcorner\,(P(\ulcorner\tau\urcorner) \to \varphi(\ulcorner\tau\urcorner))$ is used for some predicate P distinguishing a class of formal terms.

Substantive sense

Quantifiers over formal terms make possible statements of the form «every description of a property has such-and-such structure», «there exists a formal term with such-and-such characteristic». These are meta-level statements — about the syntactic language itself.

Their applicability is limited by the fact that they speak about syntax, not about objects. The statement «there exists a formal term ⌜dragon⌝» is true trivially (dragon is described); the statement «there exists a dragon» in the sense of being is a formula of the operational register, and its truth is determined by the presence of an operational realisation (of which there is none).

VII.4. The law of the excluded middle

Status of LEM in the operational register

In the operational register of VR-Sets the law of the excluded middle $(\varphi \vee \neg\varphi)$ holds — an inheritance from the classical logic of VR-Sets. Substantively: every operational functionality gives a definite answer; there is no «middle».

Status of LEM in the formal register

In the formal register LEM also holds — since the logic is classical. However, its substantive significance requires caution.

The statement «⌜dragon⌝ exists ∨ ¬(⌜dragon⌝ exists)» is true in the formal register by LEM. But «exists» here is formal (i.e. «the formal term ⌜dragon⌝ is»), not a claim of being. So the statement is trivial: the formal term ⌜dragon⌝, of course, is (we have recorded it); hence the disjunction is true, since its first part is true.

When someone says «a dragon exists or does not exist» in the sense of being — this is in another register. In the operational register the formula «∃x dragon(x)» translates into a statement about the existence of an operational realisation of a dragon; its truth value is determined by operational realisation (and in this case is false — no operational realisation).

Remark

The distinction of registers in the application of LEM removes many classical philosophical difficulties. «Does god exist?» in the formal register — yes, ⌜god⌝ is a formal term. In the operational — no, no operational realisation in the operational universe of VR. This is not a contradiction but a difference of semantic registers of the question.

VR-Forms does not claim that one of the answers is «more correct» than the other. It asserts that they answer **different questions**, and explicit distinction of registers helps not to confuse them.

VII.5. Connection with other formal systems

Let us collect the explicit comparisons of VR-Forms with systems known in logic.

Two-sorted logic

A standard extension of first-order logic in which variables are divided into sorts. VR-Forms is formally a two-sorted system (operational sort, formal sort). Peculiarity: the formal sort includes the operational (every operational set has a formal correlate), but not vice versa. This is an asymmetric embedding.

NBG (Bernays–Gödel)

An extension of ZFC with classes as second-sort objects. VR-Forms is structurally analogous: formal terms play the role of classes in NBG (large collections that are not operational sets). Difference: classes of NBG are objects (although of the second sort); formal terms of VR-Forms are formulas. This gives VR-Forms greater clarity regarding status: the formal register makes no claim of being.

Internal Set Theory (Nelson)

An extension of ZFC separating standard and nonstandard objects. Structural similarity with VR-Forms: two registers, passage between them. Difference in motivation: IST introduces the nonstandard as a tool for proofs about the standard; VR-Forms introduces the formal as a way to speak about the non-existent without committing to its being.

Furthermore: IST extends ZFC with new objects that «are» in some sense; VR-Forms adds no objects to VR-Sets (formal terms are formulas, not objects). This is a principial distinction in the spirit of the position.

Constructive and intuitionistic logic

Constructivism (Brouwer, Bishop) and intuitionism (Heyting) reject LEM and non-reflexive proofs of existence. VR-Forms preserves classical logic. The difference is principled: constructivism refuses language for the non-constructive; VR-Forms permits the language but deprives it of operational force.

One could say: VR-Forms gives classical logic in the formal register and operations in the operational register. This removes the tension between «classical mathematics works» and «not all classical objects are real»: they work as formulas; real are those that are operationally realisable.

Realisability (Kleene, Kreisel)

Realisability connects classical and constructive mathematics: a formula is classically provable and constructively realisable if it has a constructive witness. The notion of operational realisability (Definition II.3) is an analogue of realisability in VR-Forms. The difference: realisability classically answers «can the proof be constructivised?»; operational realisability in VR-Forms answers «does the formal term have an operational correlate?»

Free logic (Lambert, Bencivenga)

Free logic admits terms not having reference (terms denoting the non-existent). VR-Forms is formally similar to free logic: formal terms are terms without reference. Difference: free logic is a single theory with a single register; VR-Forms has two registers, with strictly formalised passage between them.

VII.6. Summary of Part VII

Established:

(1) Classical first-order logic acts in both registers of VR-Forms (§VII.1).

(2) Rules for working with mixed formulas; the translation π applies to them naturally (§VII.2).

(3) Quantifiers over formal terms are quantifiers over a countable set of syntactic objects; do not generate uncountability (§VII.3).

(4) The law of the excluded middle holds in both registers; careful distinction of registers removes many classical difficulties with LEM for non-operational objects (§VII.4).

(5) Connections of VR-Forms with two-sorted logic, NBG, IST, constructivism, realisability, free logic (§VII.5).

Methodological summary. VR-Forms is logically closed. Its inference rules are clear, the relations between registers are formalised, connections with known systems are established. The system makes no hidden logical assumptions beyond classical first-order logic and the principles of VR-Sets.

What follows. Part VIII — the concluding part — records open questions, indicates the programme of VR-Audit as a natural continuation, and gives the general summary of the VR cycle.

Part VIII. Open Questions and the Place of VR-Forms in the Cycle

VIII.0. Introduction

The concluding part of the preprint sums up VR-Forms and records open directions. The structure parallels Part IX of VR-Sets: technical questions, substantive extensions, philosophical open directions, general conclusion.

Part VIII closes the work. All formal results are gathered in Parts II–IV; applications in Parts V–VI; logical properties in Part VII. Here — the fixing of what remains open and of what is planned as the next work of the cycle (VR-Audit).

VIII.1. Technical open questions

Question 1: formalisation in Lean/Coq/Agda

Machine formalisation of VR-Forms is a natural technical step. The two-register structure fits well with dependent types: the operational register — usual types with values, the formal register — syntactic objects with an interpreting mapping. Lean 4 appears most suitable: its mathematical library (mathlib) already contains tools for working with conservative extensions, translations, two-sorted logic.

Concrete formalisation tasks: (a) formal verification of Theorem III.1 on conservativity; (b) formalisation of the transit rule (§IV.2); (c) machine verification of examples (§IV.5, Parts V–VI).

Question 2: precise strength of the translation π

The mapping π from §III.2 translates formal terms into defining predicates. On simple formal terms of the form $\ulcorner\{x : \varphi(x)\}\urcorner$ the translation is trivial. For more complex constructions (quantification over formal terms, iteration of formal terms, recursive definitions via formal terms) the translation requires more careful definition.

Open question: what is the precise range of formal constructions for which π gives an invertible translation? Are there formal terms for which π gives an ambiguous result? These questions matter for a strict variant of formalisation in Lean.

Question 3: composition of transits

If there is a transit D_1 yielding the operational statement φ, and a transit D_2 using φ as a lemma on the way to a statement ψ — is there a single transit for ψ, or is a composition of separate transits needed?

This is a question of practical work with VR-Forms: how naturally do chains of lemmas organise into one large transit, and do problems with conservativity arise under composition?

It is expected that the composition of transits is safe (by transitivity of conservativity), but a strict proof and a survey of practical cases remain open.

Question 4: alternative logics of the formal register

In §VII.1 the choice was made in favour of classical logic in the formal register. Open question: what properties of VR-Forms are preserved, and which change, under replacement of classical logic with:

(a) intuitionistic logic — substantively: the formal register becomes constructive; paradoxical formal terms receive a different status;

(b) paraconsistent logic — substantively: contradictions in formal terms may be treated explicitly, without threat of overall contradiction;

(c) modal logic — substantively: formal terms may differ in «mode» (necessary/possible/counterfactual).

Each of these alternatives is a separate work.

VIII.2. Substantive extensions

Extension 1: VR-Audit

The direct and principal continuation of VR-Forms is the programme of VR-Audit. Content: systematic analysis of classical mathematical theorems with the aim of determining which of them transit directly into VR-Sets, which require operational labour to extract content (problematic transits), and which are essentially non-operational.

VR-Forms provides the apparatus for VR-Audit. VR-Audit uses this apparatus on concrete theorems. List of first candidates:

— the spectral theorem in its various formulations (discrete and continuous spectrum);

— the Hahn–Banach theorem in general form;

— Baire's category theorem;

— Tikhonov's compactness theorem;

— the Radon–Nikodym theorem on measures;

— theorems from quantum mechanics (Stone, von Neumann, Naimark).

Each such theorem can be analysed: is the uncountable used essentially, or only as convenience?

Extension 2: connection with reverse mathematics

The programme of Reverse Mathematics (Friedman, Simpson) classifies classical theorems by the strength of axioms needed for their proof. The hierarchy of subsystems of second-order arithmetic (RCA_0, WKL_0, ACA_0, ATR_0, Π^1_1-CA_0) gives a precise measure of the «uncountable strength» of each theorem.

VR-Forms and VR-Audit can connect with this programme through the translation π. Each system of the Reverse Mathematics hierarchy corresponds to a definite class of formal terms in VR-Forms; theorems provable in RCA_0 transit directly into VR-Sets; theorems requiring stronger subsystems yield problematic transits, and the character of problematicity depends on the subsystem.

Establishing the precise correspondence is a separate task, uniting VR-Forms with the already substantial Reverse Mathematics literature.

Extension 3: connection with topoi and category theory

Categorical set theory (Lawvere ETCS, Macnamara SDG, elementary topoi) gives an alternative view of foundations through universal properties. VR-Sets is close to them in spirit (see VR-Sets §IX.2). VR-Forms adds the possibility of speaking about non-operational categorical constructions as formal terms: ⌜topos of all functors⌝, ⌜absolutely complete category⌝, etc.

Open direction: construction of VR-categorical apparatus, where operational categories (countable, with countable structure) are operational, and larger ones are formal.

Extension 4: homotopy type theory (HoTT)

In HoTT equality is not a relation but a type (path). VR-Forms can be reinterpreted in the HoTT spirit: ≡ as a path in the type of operational sets, formal terms as higher types with a special interpreting translation. This is technically complex but substantively close.

Extension 5: operational foundations of physics

In VR-Sets §IX.3 the question of the applicability of the operational approach to quantum mechanics was raised. VR-Forms provides a more complete apparatus for this direction.

Hypothesis: quantum mechanics of operational systems (with discrete spectrum of the Hamiltonian) is fully formulated in VR-Sets; quantum mechanics of continuous systems requires the formal register for its uncountable constructions (classical Hilbert space, spectral measure, projection-valued measure). At the same time the physically observable results — measurable quantities — should transit back into the operational.

Verification of this hypothesis on concrete physical examples is a task leading to a possible future work VR-Physics.

VIII.3. Philosophical open questions

Question 1: status of the formal term

In Part I §I.4 it was fixed: a formal term is simply a formula, without any claim of being. This position is consistent but philosophically rigid. Open question: can an intermediate status of formal terms be found — for example, «existing in the sense of use», «existing as linguistic objects», «existing in Popper's third world»?

The present work declined such intermediate positions for the sake of clarity. The alternatives remain open for investigation.

Question 2: place of VR-Forms among the foundations

VR-Forms positions itself as a minimalist operational foundation with formal extension. Close positions: Feferman's predicativism (FOM community), Bishop's countable constructivism, Bridgman's operationalism. VR-Forms differs from each, but the precise place on this map requires a separate philosophical study.

Question 3: the next work of the cycle

VR-Forms is the fourth work of the cycle. Open candidates for the fifth:

(a) **VR-Audit** — systematic analysis of classical mathematics through the lens of VR-Forms;

(b) **VR-Categories** — categorical reformulation of VR-Sets and VR-Forms;

(c) **VR-Physics** — operational reformulation of the foundations of physics;

(d) **VR-Logic** — detailed elaboration of the A1-duality into a self-standing operational logic.

The natural choice is VR-Audit, as a direct application of the VR-Forms apparatus. VR-Categories and VR-Physics presuppose more extended programmes. VR-Logic is a parallel branch developed independently.

VIII.4. Conclusion

Completion of the VR cycle

The VR cycle in its current form — four works:

(1) **VR. A Formal System** — arithmetic on a no-ontology base.

(2) **VR-Numbers** — numerical extensions on the same base.

(3) **VR-Sets** — set theory; for the first time reveals the boundary of the operational (countability of the universe).

(4) **VR-Forms** — a formal language for speech about what lies beyond the boundary. Closes the operational contour: now VR can speak about everything, ascribing no ontology — only operations, ∅ being the base operation.

The slogan of the cycle, carried through the four works:

«Nothing simply is. Everything either acts or is said. What acts is operational, with ∅ the base operation; what is merely said is formal and is backed by no operation.»

What the cycle demonstrates

The VR cycle in its completed form demonstrates: **a no-ontology foundation is possible and workable. On the base operation ∅ are built** arithmetic, numerical extensions, set theory; for the description of the non-operational a formal register is introduced, not violating the minimalism. No operational commitments are accepted; nothing of classical mathematics or speech in general is left without a place.

This is a philosophical position carried through a formal apparatus. It does not claim exclusivity: classical ZFC, constructivism, formalism, categorical set theory — all remain legitimate alternatives. But the VR cycle shows that a minimal operational foundation is a substantive **and consistent option**, capable of serving as a foundation for mathematics and as a language for speech in general.

What the cycle does not do

The cycle does not prove that ∅ «is» — ∅ is taken as a nullary operation, a starting stance, not a theorem. The cycle does not deny other ontological positions — it only clearly fixes its own and works relative to it. The cycle does not claim instrumental superiority over classical foundations — the programme of VR-Audit is precisely what is intended to systematically check in which cases VR provides something beyond the known.

The mature role of VR is not as an alternative to classical foundations, but as a **transparent and minimalist point of view**, relative to which classical foundations receive a clear classification (through VR-Audit). This is methodological value, not mathematical revolution.

Part IX. Lean 4 Formalisation of VR-Forms: Methodological Observations

IX.0. Introduction

The present part documents the Lean 4 formalisation of the two-register apparatus of VR-Forms (Parts II–IV and §VII.2 of the present preprint), carried out after the publication of v1.0.0 and presented as a self-contained companion software publication (Reznik 2026, Lean VR-Forms, Software DOI 10.5281/zenodo.20355757; Git tag v1.3-vr-forms). The formalisation realises the apparatus as a shallow embedding over mathlib and surfaces a single explicit structural boundary at conservativity (Theorem III.1).

Part IX is the fourth such methodological-observations part in the VR cycle, following Part VIII of VR-Numbers v1.0.2 and Part X of VR-Sets v1.0.1. Its function is the same: not to introduce new mathematical content, but to record what the formalisation made visible — observations about the structural relationship between the preprint's apparatus and Lean's proof-theoretic infrastructure.

Ten observations are presented in four thematic clusters. §IX.1 collects two observations on foundation-level properties of the formalisation. §IX.2 collects five observations centred on the central structural boundary of the cycle (conservativity) and its four substructural manifestations. §IX.3 collects two observations on structural patterns in the apparatus. §IX.4 closes with two cross-cycle observations, including the finale on zero Classical.choice usage. §IX.5 summarises the comparison with the boundaries documented in VR-Sets Part X.

Methodologically, Part IX resolves Question 1 of §VIII.1 («Formalisation in Lean/Coq/Agda») of the present preprint for the formalisable core (the two-register apparatus); the central boundary at conservativity is documented but not crossed. The Lean formalisation is also accessible directly: each observation has a corresponding source location in the Lean code documented in comment form.

IX.1. Foundation-level properties

Observation 1: foundation file is import-free and axiom-free

The foundation file `Forms/Language.lean` introduces the syntactic skeleton of the apparatus (the `Register` inductive type, the `FormalTerm` structure, the notation `⌜·⌝`) without importing any mathlib or VR-Sets module. `String` and inductive/structure constructions are part of the Lean 4 prelude. The axiom profile is empty `[]` for both `Register` and `FormalTerm`.

This contrasts with the foundation files of VR-Numbers and VR-Sets Lean. `Numbers/Foundation.lean` imports mathlib's natural number infrastructure; `Sets/Foundation.lean` imports mathlib's `ZFSet` and inherits `[propext, Quot.sound]` for every theorem. The contrast reflects the deliberate architectural choice of shallow embedding in VR-Forms: formal terms are syntactic metadata, not mathematical objects in mathlib's hierarchy.

The methodological consequence is structural. The base layer of VR-Forms is the most axiom-minimal foundation of the four cycles; any axiom dependency in subsequent stages arrives **through** the connection to VR-Sets (which carries `propext` and `Quot.sound` from the ZFSet quotient), not through the formal-term apparatus itself.

Observation 2: realisability inherits the Classical-free closure layer of VR-Sets

The predicate `isRealisable : FormalTerm → Prop` is defined in `Forms/Realisability.lean` via `match` on `FormalTerm`, with cases for the realisable terms `⌜∅⌝`, `⌜omega_OSet⌝`, `⌜osetPair⌝`, plus VR-Sets-refutable `⌜AFA_Statement⌝`, plus the open Conjecture cases (added retroactively in Stage 4), plus a catch-all `False`. All four base realisability lemmas sit at `[propext, Quot.sound]`, with **no Classical.choice**.

The structural reason is direct. The realisable cases (∅, ω, pair) correspond to Theorems III.1–III.3 and Theorem III.6 of VR-Sets, which are Classical-free closure theorems. The realisability layer of VR-Forms is structurally tied to the Classical-free closure layer of VR-Sets — realisability of operationally well-defined terms inherits exactly the axiom dependency of their closure proofs.

Realisable terms tied to *Classical* VR-Sets theorems (replacement, choice, foundation) would inherit `Classical.choice`. No such terms appear in the present formalisation; this is consistent with the preprint's position that the operationally well-defined VR objects (∅, ω, pair) form the cycle's realisable core, while constructions essentially using Classical infrastructure remain formal terms without realisation.

IX.2. The central boundary and its substructural manifestations

The defining feature of VR-Forms Lean is its **single explicit structural boundary** at conservativity. This contrasts with VR-Sets Lean, which surfaced five distinct boundaries between the operational universe and mathlib's set-theoretic infrastructure. The VR-Forms boundary is methodologically different: it sits at the proof-theoretic meta-level (between shallow and deep embedding), not at the mathematical-content level (between operational and classical sets).

§IX.2 documents the central boundary (one observation) and four substructural manifestations encountered in the formalisation (four observations).

Observation 3: conservativity (Theorem III.1) is the explicit structural boundary

Theorem III.1 of the present preprint (§III.2) — the conservativity of T_1 over T_0 in the operational register — is **not** formalised in the Lean cycle. Instead, it is documented as the explicit structural boundary, with verbatim citation in the doc-comment of `Forms/Transit.lean`.

The reason is architectural. Full formalisation of conservativity would require deep-embedded `Formula` L_1, `Derivation` T_0, `Derivation` T_1 types, the translation π as a function on the formula syntax, and an induction proof over derivations. This is a proof-theory project larger than the entire VR-Sets Lean cycle, and would shift the formalisation's role from «Lean library reflecting the preprint» to «Lean library *about* proof theory». The decision is documented in the Lean cycle's `CLAUDE.md` and `PLAN.md`, with three alternatives (full deep embedding, shallow embedding making conservativity trivial, partial formalisation with explicit boundary) discussed before the cycle began.

The methodological position is honest: conservativity is **mathematically proved** in the preprint (§III.2 gives the full inductive proof via the translation π), but **Lean-unformalisable** at the shallow-embedding depth chosen here. This is distinct from open conjectures, which are mathematically open rather than formalisation-limited. The Lean cycle records this distinction by *not* introducing a `def Conjecture_Conservativity : Prop` — such a definition would misrepresent the result as open to Lean attack, when in fact it is proved (just outside the shallow apparatus).

The transit pattern (§IV.2) is documented as an inference template in `Forms/Transit.lean`, not formalised as a Lean theorem. The conservativity justification flows through external reference to the preprint's §III.2 proof.

Observation 4: equation-compiler catch-all does not reduce in term mode

The first substructural manifestation of the boundary is technical. The function `translate_pi : FormalTerm → Prop`, defined via `match` with named cases plus a catch-all `| _ => False`, does **not** reduce `translate_pi x` to `False` in term mode for an arbitrary `x : FormalTerm`. The reduction is blocked by the equation compiler's schematic-variable representation of the catch-all: the term `translate_pi x✝` (where `x✝` is the catch-all's schematic variable) stays unreduced even when the user knows that `x` is not one of the named cases.

The working proof structure (used in `translate_implies_realisable`) is `by_cases` discrimination on the named constructors — which `DecidableEq FormalTerm` makes possible without Classical.choice — followed by `unfold translate_pi; split <;> simp_all [FormalTerm.mk.injEq]` in the residual catch-all branch. The fix costs zero new axioms but requires explicit case structure.

The observation parallels the proof technique in VR-Sets Stage 8, where `Classical.epsilon` had to be applied with `eq_empty_or_nonempty` case-split to make reduction work. Both cases share the same structural fact: Lean's pattern matching requires explicit case discrimination at the boundary between abstract scrutinee and concrete reduction.

Observation 5: realisability and π-translation form complementary layers

The second substructural manifestation: `isRealisable` (existential — «term has operational witness») and `translate_pi` (specific — universal-quantified statement about a named VR-Sets object) form two complementary layers in the Lean apparatus, connected by `translate_implies_realisable : ∀ t, translate_pi t → isRealisable t` via existential introduction.

The preprint §II.7 / §III.2 conflates the two notions (operational realisability and π-translation are presented as aspects of the same apparatus). The Lean formalisation separates them, and only the forward direction (specific → existential) is proved. The converse (existential → specific) is **not** provable from the existential alone: from «there exists some operational witness» one cannot extract «the specific witness `osetEmpty`» without Skolemisation, which is not available from the existential in the shallow embedding. The transit pattern of §IV.2 operates exclusively in the forward direction, which is structurally consistent.

This separation is the formal content of what the preprint calls «the transit pattern operating forward»: from an operational truth about a specific VR-Sets object (the π-translation), the realisability of the corresponding formal term follows.

Observation 6: two-level structure of negative cases

The third substructural manifestation concerns non-realisability. The Lean formalisation distinguishes **two levels** of non-realisable formal terms.

Level 1 (trivially False). For «mythological» terms — those without mathematical formulation in VR-Sets — the proof is `id`: `¬isRealisable ⌜"Vitali"⌝ := id`, `¬isRealisable ⌜"Russell_class"⌝ := id`, and so on. The catch-all `| _ => False` redirects all unnamed terms to `False`, and negation is trivial. These four theorems (Russell, Vitali, classical ℝ, classical ℘(ℕ)) are in `Forms/Examples.lean`. The proof carries no VR-Sets mathematical content; non-realisability is by *absence from the realisability list*.

Level 2 (VR-Sets-refutable). For `⌜AFA_Statement⌝`, the case is different. The match returns `VR.Sets.AFA_Statement` (a Prop with mathematical formulation in mathlib's PSet inductive structure). The bridge theorem `bridge_AFA : ¬isRealisable ⌜"AFA_Statement"⌝ := AFA_Refuted` (in `Forms/Bridge.lean`) is a direct application of `AFA_Refuted` from VR-Sets Stage 10. Non-realisability carries genuine mathematical content: AFA contradicts the well-foundedness of PSet.

The two-level structure formalises a distinction implicit in the preprint between «paradoxical descriptions without operational correlate» (Russell, Vitali) and «descriptions whose negation has independent mathematical content» (AFA). The latter category is rarer but methodologically more substantial.

Observation 7: three-category structure of formal terms

The fourth substructural manifestation. The bridge module surfaces a triadic classification of formal terms by realisability status:

— **(a) Provably realisable** — Stage 2 lemmas with concrete witnesses (`isRealisable_empty`, `isRealisable_omega`, `isRealisable_osetPair`). The realisable list is closed and explicit; each entry corresponds to a closure theorem of VR-Sets.

— **(b) Open realisability** — Conjecture formal terms (`bridge_Conjecture_IV_1`, `bridge_Conjecture_IV_2`). The bridge `iff` is trivially provable by definitional reduction, but the content (whether the conjecture holds) is mathematically open. Both directions of the `iff` reduce to `id`.

— **(c) Provably non-realisable** — split into the two levels of Observation 6.

The triadic structure parallels VR-Sets Stage 11's three-tier formalisation result (proved theorems, refuted claims, open formulations), but localised at the level of formal terms rather than at the level of mathematical claims. The bridge module thus inherits and refines the VR-Sets tier structure within the formal register of VR-Forms.

IX.3. Structural patterns

Observation 8: universe handling across cross-cycle boundaries

Two distinct universe-management issues surfaced in the formalisation; both arise from the cross-cycle nature of VR-Forms.

Issue (a) — Stage 2. In the body of `def isRealisable`, references to `OSet` require the explicit annotation `OSet.{0}`. Without it, Lean cannot infer the universe level when `OSet` is the codomain of a `match`-induced quantification. The issue is namespace-related: inside `namespace VR.Sets`, surrounding context fixes the universe; inside `namespace VR.Forms` with `open VR.Sets`, the universe must be pinned explicitly.

Issue (b) — Stage 5. Projecting `.1` from `Theorem_III_6_Infinity` (a two-universe `{u v}` signature inherited from `ZFSet/PSet` interaction) generates universe metavariables outside the defining namespace. The workaround is to use `translate_pi_omega.1` instead, whose match-case pins `OSet.{0}` and makes the projection unambiguously typed.

Both issues point to the same architectural fact: VR-Forms is structurally a *cross-cycle* module, and Lean's universe inference does not propagate cleanly across cycle boundaries without explicit annotation. This is a mathlib/Lean 4 idiosyncrasy, not a logical issue — but it costs care in implementation, and it is the first such cross-namespace technical concern in the VR Lean programme.

Observation 9: ZFA boundary manifests simultaneously in both registers

The Lean finale theorem `mixed_AFA_boundary` (in `Forms/Examples.lean`) captures, in a single Lean Prop, the parallel manifestation of well-foundedness across the two registers of VR-Forms:

```
theorem mixed_AFA_boundary :
```

```
  (∀ x : OSet.{0}, x ∉ x) ∧ ¬isRealisable ⌜"AFA_Statement"⌝ :=
⟨ZFSet.mem_irrefl, bridge_AFA⟩
```

Operationally: `∀ x : OSet, x ∉ x` — regularity, via `ZFSet.mem_irrefl`. Formally: `¬isRealisable ⌜"AFA_Statement"⌝` — non-realisability of the AFA formal term, via `bridge_AFA`. Both sides derive from the inductive nature of mathlib's `PSet`, and surface the same structural fact (well-foundedness of mathlib's set-theoretic infrastructure) through different conceptual layers.

This is the most mathematically substantive mixed formula of the VR-Forms cycle and the structural counterpart to VR-Sets's Stage 10 boundary B.5 («ZFA total absence»). What appeared in VR-Sets Part X as the deepest *structural* boundary appears in VR-Forms Lean as a *theorem* with cross-register content — the same architectural fact, now formulated in the apparatus that VR-Forms provides for talking about both registers at once.

IX.4. Cross-cycle integration

Observation 10: zero Classical.choice usage across the entire cycle

The final observation closes the cycle. The original cycle plan (`PLAN.md` of the Lean repository) predicted that the public objects of VR-Forms Lean would sit at `[propext, Classical.choice, Quot.sound]` or stricter — admitting some Classical use. The actual result, verified by `#print axioms` for every public object, is stricter than predicted:

File	Public objects	Axiom profile
Language.lean	3	`[]` empty
Realisability.lean	4	`[propext, Quot.sound]`
Transit.lean	5	`[propext, Quot.sound]`
Bridge.lean	3	`[propext, Quot.sound]`
Examples.lean	6	`[propext, Quot.sound]`

Eighteen public objects total. Three are axiom-free (Stage 1). Fifteen sit at `[propext, Quot.sound]`. **Zero objects require Classical.choice.**

This contrasts sharply with the predecessor cycles. VR Part I and VR-Numbers Lean each contain several objects depending on Classical infrastructure; VR-Sets Lean has six of twenty-two objects at the full ceiling `[propext, Classical.choice, Quot.sound]`, through four structurally distinct Classical mechanisms (ordinal-valued constructions, definability, foundation, choice).

The structural reason is direct. The realisable cases of VR-Forms correspond to the *Classical-free* closure theorems of VR-Sets (Observations 1 and 2); the non-realisable cases either reduce trivially (Russell, Vitali via `False`) or use `AFA_Refuted` (Classical-free, proved via inductive PSet reasoning). No formalisation step in VR-Forms requires a

Classical mechanism, because the formalisable apparatus operates entirely within the operationally well-behaved subset of VR-Sets.

The observation reflects the nature of the present preprint: a *formal language* for the non-operational, not a substantive mathematical theory. The Lean formalisation, when carried out at the chosen depth, lives entirely within Lean's constructive-plus-quotient core.

IX.5. Comparison with VR-Sets Part X

The structure of boundaries in VR-Forms Lean differs methodologically from VR-Sets Lean (documented in Part X §X.3 of VR-Sets v1.0.1).

VR-Sets surfaced **five** structural boundaries between the operational universe and mathlib's set-theoretic infrastructure (B.1 powerset cardinality, B.2 replacement schema, B.3 foundation modes, B.4 ZFA modal status, B.5 ZFA total absence). Each boundary is a gap in mathematical content — what the preprint claims about the operational universe versus what mathlib provides in its type-theoretic infrastructure.

VR-Forms surfaces **one** structural boundary at conservativity. The boundary is at the proof-theoretic meta-level (between shallow and deep embedding), not at the mathematical-content level. The four observations of §IX.2 are not separate boundaries but substructural manifestations of the single central boundary in different aspects of the formalisation.

The shift in structure — from five mathematical-content boundaries to one proof-theoretic boundary with four manifestations — reflects the shift in subject matter. VR-Sets formalises a set theory; its boundaries are about what mathlib's sets can and cannot do. VR-Forms formalises a *language for talking about sets*; its boundary is about what the shallow embedding can and cannot say about its own metatheory.

Both cycles arrive at axiom-minimal results, but through different routes. VR-Sets achieves axiom-minimality by *navigating* the boundaries (proving theorems despite mathlib's structural mismatches); VR-Forms achieves axiom-minimality by *staying entirely on the shallow side* of its single boundary.

IX.6. Summary of Part IX

Ten observations have been documented:

(1) Foundation file is import-free and axiom-free; the most axiom-minimal foundation of the four VR Lean cycles. (§IX.1)

(2) Realisability inherits the Classical-free closure layer of VR-Sets; no realisable case requires Classical.choice. (§IX.1)

(3) Conservativity (Theorem III.1) is the explicit structural boundary — mathematically proved in the preprint, Lean-unformalisable at shallow depth, documented in `Transit.lean`. (§IX.2)

(4) Equation-compiler catch-all does not reduce in term mode; `by_cases + unfold + split + simp_all` is the zero-axiom fix. (§IX.2)

(5) `isRealisable` (existential) and `translate_pi` (specific) form complementary layers; forward implication is the formal content of the transit pattern. (§IX.2)

(6) Two-level structure of negative cases: trivially False (Vitali, Russell) vs VR-Sets-refutable (AFA via `AFA_Refuted`). (§IX.2)

(7) Three-category structure of formal terms: provably realisable, open realisability, provably non-realisable. (§IX.2)

(8) Universe handling across cross-cycle boundaries — `OSet.{0}` annotation and `translate_pi_omega.1` projection workaround. (§IX.3)

(9) ZFA boundary manifests in both registers simultaneously; `mixed_AFA_boundary` formalises this as a single Lean Prop. (§IX.3)

(10) Zero Classical.choice usage across the entire cycle — the most axiom-minimal of the four VR Lean cycles. (§IX.4)

The full Lean source is at https://github.com/inventor1975/VRCycle (tag v1.3-vr-forms); each observation has a corresponding source location documented in Lean comments.

Methodological summary. The formalisation realises in Lean exactly the apparatus that VR-Forms is — a shallow, operationally minimal, axiom-light layer over VR-Sets, with one explicit structural boundary at the place where it would have to become a proof-theory project. The boundary is honest, the apparatus on the formalisable side is complete, and the cycle as a whole is the most axiom-minimal of the four VR Lean formalisations.

What this resolves. Question 1 of §VIII.1 («formalisation in Lean/Coq/Agda») is now answered for the formalisable core. Question 2 («precise strength of the translation π») receives a partial answer: in the shallow embedding, π is total and exact for the named realisable cases. Question 3 («composition of transits») is not addressed in the formalisation — the transit pattern is documented but not compositionally analysed. The full conservativity formalisation (Question 1 in its strict sense) remains the structural boundary.

What follows. The VR cycle in its current form — four works with four Lean formalisations — closes the planned operational contour. The next directions, sketched in §VIII.2 of the present preprint, are VR-Audit (the direct application of the present apparatus to classical mathematics), VR-Categories, and VR-Physics. The Lean cycle of VR-Forms, by realising the formalisable core of the apparatus, prepares the technical infrastructure for VR-Audit: the bridge module and the transit pattern are the working tools through which classical theorems can be systematically classified.

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The Lean 4 formalisation documented in Part IX (Reznik 2026, Lean VR-Forms, Software DOI 10.5281/zenodo.20355757) was carried out using a parent-child review architecture: Claude Opus 4.7, holding the present preprint, acted as architectural reviewer and made all stage-level design decisions; Claude Sonnet 4.6, running in Claude Code, wrote the Lean code, ran lake build, and reported #print axioms for every public object. Each of the six stages followed the protocol of plan-before-code with tested prototype, acceptance criteria,

and axiom audit. The same architecture was used in the preceding Lean cycles of VR-Sets and VR-Numbers.

Part IX itself was drafted by Claude Opus 4.7 from the methodological observations accumulated through the Lean cycle, with verbatim references to the corresponding source locations in the Lean code.

All mathematical content, definitions, theorems, philosophical positions, and architectural decisions are due to the author. Working draft 1.0.1a reframes the exposition to the cycle's no-ontology position: ∅ is taken as a nullary operation, the cycle's slogan is «nothing is — all is doing», and the two registers are named operational/formal (the earlier label «ontological register» is dropped as the cycle ascribes no ontology); the apparatus, theorems, and all results are unchanged. This revision was carried out with Claude Opus 4.8.